

Mapping Medicaid Deserts:

Provider Shortages, Preventable Hospitalizations, and the Limits of Workforce Policy in California's Medi-Cal System

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Abstract

Having health insurance does not guarantee access to healthcare. This paper studies *Medi-Cal deserts* in California: counties where many residents are enrolled in Medi-Cal but primary care doctors are in short supply. Using data from more than 20 government sources covering 2008 to 2024, I build an Access Resource Index (ARI) to rank all 58 California counties by how well-served they are, then measure how living in a desert county affects hospital admissions, emergency room use, death rates, and costs.

Desert counties have 48.6 more preventable hospital admissions per 100,000 residents than non-desert counties ($p < 0.05$), a rate 37% higher. Patients in desert counties are also 38% more likely to leave the emergency room before being treated, death rates per county run 21 to 34% higher (widening during COVID-19), and hospitals in these counties operate at a loss while non-desert hospitals earn a small profit. Telehealth use grew nearly ten times during COVID-19 and has stayed high, which may help close some of the access gap going forward.

I test five causal methods to see whether specific programs (National Health Service Corps (NHSC), Federally Qualified Health Center (FQHC), Rural Health Clinic (RHC), Steven M. Thompson Loan Repayment Program (STLRP)) reduce preventable hospitalizations. None find a statistically significant effect. This does not mean the programs are failing. It means the problem is structural and larger than any single program can fix on its own, and that with only 58 counties the analysis lacks the statistical power needed to detect modest effects. Programs are targeting the right counties. They are just not large enough to overcome the deeper barriers driving the desert problem: low Medi-Cal payment rates, financially struggling hospitals, lack of transportation, and concentrated poverty.

The estimated annual excess cost from poor access in desert counties is \$82 million. Fixing this will require a set of reforms working together: raising Medi-Cal payment rates, expanding loan repayment programs, allowing nurse practitioners to practice independently, building rural training pipelines, investing in telehealth, supporting safety-net hospitals financially, and improving transportation to medical appointments.

Keywords: Medi-Cal, access deserts, primary care, workforce redistribution, Prevention Quality Indicators, NHSC, STLRP, Proposition 56, Medicaid policy, California, healthcare equity

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PART I

The Problem

*Why insurance alone cannot guarantee care, and what Medi-Cal deserts reveal
about California's healthcare access crisis*

1. Introduction

California expanded Medi-Cal under the Affordable Care Act, growing enrollment from about 8 million to over 14 million people by 2024. More than one in three Californians is now covered by Medi-Cal. But having insurance does not automatically mean getting care. Medi-Cal patients still face long wait times, difficulty finding doctors who accept their insurance, and heavy reliance on emergency rooms for conditions that a primary care visit could handle.

This paper asks why. The answer is not mainly about how Medi-Cal is designed or administered. California enrolls nearly all Medi-Cal patients in managed care plans, and those plans function reasonably well in most of the state. The deeper problem is that primary care doctors are not evenly distributed. The Health Resources and Services Administration's (HRSA) 2024 State of the Primary Care Workforce Report projects that demand for primary care services will outstrip supply through at least 2035, with rural and high-Medi-Cal communities facing the steepest deficits. In some California counties, particularly in the Central Valley, rural Northern California, and near the southern border, there are simply not enough doctors to serve the people who need them. When you combine that doctor shortage with high Medi-Cal enrollment, financially strained hospitals, limited public transportation, and widespread poverty, you get a self-reinforcing cycle of poor access and poor health outcomes. This paper calls these places *Medi-Cal deserts*.

The consequences are measurable. Desert counties have significantly more preventable hospitalizations, higher rates of patients walking out of emergency rooms without being treated, higher death rates, and hospitals that are barely breaking even financially. Based on the analysis in this paper, desert conditions generate roughly \$82 million in excess healthcare costs every year.

California already operates several programs intended to bring doctors to underserved areas: the federal National Health Service Corps (NHSC), the state's Steven M. Thompson Loan Repayment Program (STLRP), Federally Qualified Health Centers (FQHCs), Rural Health

Clinics (RHCs), the Song-Brown workforce training program, and Proposition 56, which raised Medi-Cal payment rates starting in 2017. This paper evaluates each of these using multiple statistical methods, being transparent about what the data can and cannot tell us. The short answer is that these programs are pointing in the right direction and targeting the right counties, but they are not large enough to overcome the size of the underlying problem. The policy recommendations that follow go beyond just expanding existing programs. What is needed is a broader strategy that addresses the root causes of desert conditions: payment rates that make it financially difficult to practice in low-income areas, hospitals that cannot stay solvent, training pipelines that do not include geographic requirements, and barriers like transportation that prevent patients from reaching care even when providers exist.

2. Research Questions

This paper addresses six questions:

- RQ1. Where are Medi-Cal deserts in California, and how severe are they?** I build and apply the Access Resource Index to classify all 58 counties and map where access is worst.
- RQ2. Do desert counties have worse health outcomes?** I test whether desert county residents have more preventable hospitalizations, more emergency room use, higher Left Against Medical Advice (LAMA) rates, and higher mortality.
- RQ3. What does living in a desert county cost the healthcare system?** I estimate the extra spending generated by inadequate primary care access in desert counties.
- RQ4. Which policies are most associated with better outcomes?** I examine payment rates, managed care structure, workforce programs, and telehealth use.
- RQ5. Did Proposition 56 improve outcomes in high-Medi-Cal counties?** I use difference-in-differences and synthetic control methods to test whether the 2017 pay-

ment increase produced better health outcomes.

RQ6. Do workforce redistribution programs reduce preventable hospitalizations?

I use regression discontinuity, first-differences, instrumental variables, and difference-in-differences to estimate program effects.

PART II

Data and Methods

How Medi-Cal deserts are defined, what outcomes are measured, and the causal strategies used to evaluate intervention effects

3. Data and Methods

3.1 Data Sources

This paper uses more than 20 government datasets covering 2008 to 2024. Table 1 lists the main sources.

Table 1: Primary Data Sources

Dataset	Source	Years	Key Variables
PQI Rates	AHRQ/OSHPD	2005– 2024	Preventable hosp. rate
ED Encounters	HCAI/OSHPD	2008– 2024	ED visits, LAMA rate
Hospital Financials	HCAI	2012– 2017	Margins, revenue share
Death Profiles	CDPH	2014– 2023	County mortality rates
Telehealth	CA HPD	2018– 2023	Telehealth procedures
Capitation Rates	DHCS	2019– 2023	Per-member per-month
HEDIS Quality	CHCF	2009– 2018	Diabetes, prenatal care
FQHC/NHSC/HPSA/HRSA/AHRF		2022– 2024	Provider counts, sites
STLRP Recipients	HCAI	2014– 2015	County-level awards
Income/Transport	Census ACS	2006– 2010	Income, transit access
Medi-Cal Enrollment	DHCS	2014– 2025	Enrollment, share of cost
Physician Supply	HCAI	2024	PCPs per 100k

Abbreviations: AHRQ = Agency for Healthcare Research and Quality; OSHPD = Office of Statewide Health Planning and Development; HCAI = California Health Care Access and Information; CDPH = California Dept. of Public Health; HPD = Healthcare Payments Database; DHCS = Dept. of Health Care Services; HEDIS = Healthcare Effectiveness Data and Information Set; CHCF = California Health Care Foundation; HRSA = Health Resources and Services Administration; AHRF = Area Health Resources Files; FQHC = Federally Qualified Health Center; NHSC = National Health Service Corps; HPSA = Health Professional Shortage Area; STLRP = Steven M. Thompson Loan Repayment Program; ACS = American Community Survey.

3.2 Classifying Desert Counties: The Access Resource Index

3.2.1 How the Index Is Built

To study Medi-Cal deserts, the first step is to define them. The challenge is that no single official measure captures whether a county has enough primary care capacity relative to its Medi-Cal population. The federal government designates Health Professional Shortage Areas (HPSAs) and Medically Underserved Areas (MUAs), but these designations are assigned at the sub-county level, are updated irregularly, and were not designed to make systematic comparisons across all 58 California counties at once. Using raw physician counts per capita is simpler but ignores the degree of formal federal shortage recognition, which reflects additional factors like poverty rates and travel times. Neither measure alone tells the full story.

To address this, I construct an Access Resource Index (ARI) specifically for this analysis. The ARI is a composite measure that combines two complementary data sources into a single county-level score. The first component is primary care physician density: the number of primary care doctors per 10,000 county residents, drawn from the California Health Care Access and Information (HCAI) physician supply dataset. This captures the raw supply of providers available to serve the local population. The second component is a standardized shortage score derived from the federal Primary Care Service Area (PCSA) designation, which reflects not just physician counts but also federal assessments of whether a community's healthcare needs are being met, accounting for factors like geographic access barriers and the population's health needs. Both components are converted to z-scores so they are on the same scale, then averaged:

$$\text{ARI}_c = \frac{z(\text{PCP density}_c) + z(\text{PCSA shortage}_c)}{2} \quad (1)$$

The reason for averaging rather than using either measure alone is that each captures something different. A county could have a technically adequate physician count but still be

federally designated as a shortage area because most of those doctors do not accept Medi-Cal, or because the population is geographically dispersed. Conversely, a county might have a high shortage score but a relatively thin absolute physician base. Taking the average of both gives a more complete and robust picture of how underserved a county truly is.

Higher ARI scores indicate better access. Counties in the bottom 20% of the ARI distribution are classified as desert counties. This bottom-quintile threshold identifies 11 counties with consistently poor scores on both components, meaning they are genuinely disadvantaged by both measures, not just one.

One important design feature: the ARI is built entirely from doctor supply data, not from health outcome data. Preventable hospitalization rates, emergency room use, and mortality are not used to define desert status. This matters because it means the analysis is not circular. When the results later show that desert counties have worse health outcomes, that is a real empirical finding, not a mathematical artifact of how deserts were defined.

3.2.2 Which Counties Are Desert Counties

The 11 desert counties are Tulare, Tehama, Fresno, Shasta, Kern, Imperial, Siskiyou, Glenn, Modoc, Yuba, and Lake. They are mostly in the Central Valley and rural Northern California. These are among California's poorest counties, with large agricultural workforces and high Medi-Cal enrollment. Figure 1 and Table 2 in the Results section show the geographic distribution and county-level characteristics.

3.3 What Outcomes Are Measured

The main outcome is the Prevention Quality Indicator (PQI) rate: the number of preventable hospital admissions per 100,000 Medi-Cal patients. The PQI is a validated federal measure developed by the Agency for Healthcare Research and Quality (AHRQ) that counts hospital admissions for conditions like diabetes, heart failure, and asthma that should be manageable with good outpatient care. A higher PQI means more preventable hospitalizations, which

means worse primary care access.

Secondary outcomes include: emergency department (ED) visit rates; the Left Against Medical Advice (LAMA) rate (the share of ED patients who leave before being treated); county-level death rates from the California Department of Public Health (CDPH); hospital operating margins; and estimated excess healthcare costs.

3.4 Regression Models

One key technical point: the ARI score is built from 2023 to 2024 data, so it does not change over time. If the analysis used county fixed effects, which control for stable differences between counties, the desert variable would be completely absorbed and nothing could be estimated. Instead, all models include year fixed effects to account for trends affecting all counties simultaneously, and standard errors are grouped (clustered) by county.

3.4.1 Model 1: Does Desert Status Predict Higher PQI?

$$\text{PQI}_{it} = \beta_0 + \beta_1 \cdot \text{Desert}_i + \gamma_t + \varepsilon_{it} \quad (2)$$

Here Desert_i equals 1 if county i is in the bottom ARI quintile, and γ_t are year fixed effects. The coefficient β_1 shows the average difference in PQI between desert and non-desert counties, after accounting for year-level trends. This is a descriptive estimate, not a causal one.

3.4.2 Model 2: Does Income Predict PQI Even After Controlling for Desert Status?

$$\text{PQI}_i = \beta_0 + \beta_1 \cdot \text{MedianIncome}_i + \beta_2 \cdot \text{Gini}_i + \beta_3 \cdot \text{Desert}_i + \varepsilon_i \quad (3)$$

The Gini index measures income inequality within a county. A negative and significant β_1 means poorer counties have more preventable hospitalizations, even after controlling for

whether a county is a desert.

3.4.3 Model 3: Do Individual Programs Predict PQI?

For each redistribution program P :

$$\text{PQI}_i = \beta_0 + \beta_1 \cdot P_i + \beta_2 \cdot \text{Desert}_i + \varepsilon_i \quad (4)$$

These estimates will be biased in a predictable way. Programs are intentionally sent to counties with the worst outcomes, so counties with more programs appear to have worse health in the raw data. This does not mean the programs are harmful. It means they go where the need is greatest. This is called selection bias, and it is the reason the analysis uses the causal methods described below.

3.5 Causal Identification Methods

Five different causal methods are used. Each makes different assumptions and has different strengths and weaknesses. When multiple methods point in the same direction, confidence in the conclusion increases.

3.5.1 Method 1: Regression Discontinuity

The idea. Some programs require a county to have a Health Professional Shortage Area (HPSA) score above a cutoff of 14 to be eligible. Counties just above the cutoff receive funding; counties just below do not. If the analysis is restricted to counties very close to that cutoff, the two groups should be nearly identical except for program eligibility. This is similar to a natural experiment.

The model. Let S_i be a county's HPSA score and $c = 14$ the cutoff. The centered score is $\tilde{S}_i = S_i - c$, and the treatment indicator is $T_i = 1$ if the county is at or above the cutoff. The regression is:

$$\text{PQI}_i = \alpha + \tau \cdot T_i + \beta \cdot \tilde{S}_i + \gamma \cdot (T_i \times \tilde{S}_i) + \varepsilon_i \quad (5)$$

The coefficient τ estimates the jump in PQI at the cutoff. Different bandwidth restrictions (counties within 3, 5, or 7 points of the cutoff) are tested for robustness.

The limitation. The HPSA score does not create a perfectly sharp eligibility cutoff, and with only 58 counties, restricting to a narrow band near the threshold leaves only 10 to 15 counties in the analysis. This makes precise estimates very difficult.

3.5.2 Method 2: First-Differences

The idea. Rather than comparing different counties to each other (which is confounded by many permanent differences between them), first-differences compares each county to itself before and after it gained a program. By looking at change within counties over time, stable county characteristics cancel out, whether measured or not.

The model. For a county observed before (t_0) and after (t_1) gaining a program:

$$\Delta \text{PQI}_i = \text{PQI}_{i,t_1} - \text{PQI}_{i,t_0} = \alpha + \tau \cdot \Delta P_i + \varepsilon_i \quad (6)$$

where ΔP_i equals 1 if the county gained a program between the two periods. The coefficient τ estimates how much more (or less) PQI improved in counties that gained programs compared to counties that did not. The analysis uses 2015 as the before period and 2019 as the after period, avoiding the COVID-19 years.

The limitation. This method removes fixed county differences, but it cannot remove things that changed at the same time the programs started and that also affect PQI, such as new hospitals opening or economic changes in the same period.

3.5.3 Method 3: Instrumental Variables

The idea. Instrumental variables (IV) solve the selection bias problem by finding a third variable, called the instrument, that predicts which counties get programs but does not directly affect health outcomes. The instrument used here is rural population share. Rural counties receive more redistribution programs (this is the relevance condition), and the assumption is that being rural mainly affects health through provider access rather than through other channels (this is the exclusion restriction).

Two-stage estimation.

First stage: Predict program placement from rural share:

$$P_i = \pi_0 + \pi_1 \cdot \text{RuralShare}_i + \nu_i \quad (7)$$

Second stage: Use predicted placement to estimate the effect on PQI:

$$\text{PQI}_i = \beta_0 + \tau \cdot \hat{P}_i + \varepsilon_i \quad (8)$$

The first-stage F-statistic measures how strongly the instrument predicts program placement. Values below 10 indicate a weak instrument that makes the estimate unreliable.

The limitation. The exclusion restriction is likely violated here. Rural counties are poorer, have different occupational hazards, and have less access to specialty care for reasons unrelated to primary care supply. The IV results are therefore treated as exploratory and not used to drive policy conclusions.

3.5.4 Method 4: Difference-in-Differences

The idea. Difference-in-differences (DiD) compares the change in outcomes for counties expected to benefit from a policy to the change in outcomes for counties that should not

benefit, measuring this before and after the policy takes effect. Any trend that affects both groups equally cancels out, leaving only the policy effect. The key assumption is that both groups would have followed similar trends without the policy, called the parallel trends assumption.

Application to Proposition 56. Proposition 56 raised Medi-Cal payment rates starting in 2017. Counties with a higher share of Medi-Cal patients should benefit more. The treatment group is counties where Medi-Cal covers at least 40% of patients; the control group is counties below that threshold.

$$\text{PQI}_{it} = \beta_0 + \beta_1 \text{Post}_t + \beta_2 \text{HighMC}_i + \tau(\text{Post}_t \times \text{HighMC}_i) + \gamma_t + \varepsilon_{it} \quad (9)$$

The coefficient τ is the DiD estimate. A negative, significant τ would indicate that after Prop 56, high-Medi-Cal counties improved more than low-Medi-Cal counties.

3.5.5 Method 5: Synthetic Control

The idea. Synthetic control builds a statistical stand-in for the treated group (high-Medi-Cal counties) by weighting data from untreated counties (low-Medi-Cal counties). The weights are chosen so that the synthetic control matches the treated group's outcome trajectory as closely as possible before the policy took effect. After the policy, actual outcomes are compared to what the synthetic control predicts they would have been without the policy.

The optimization. Weights w_j for each control county j are chosen by solving:

$$\min_W \sum_{t < 2017} \left(\bar{Y}_{\text{treated},t} - \sum_j w_j Y_{jt} \right)^2 \quad \text{subject to} \quad \sum_j w_j = 1, w_j \geq 0 \quad (10)$$

The estimated treatment effect at each post-2017 period t is:

$$\hat{\tau}_t = \bar{Y}_{\text{treated},t} - \sum_j \hat{w}_j Y_{jt} \quad (11)$$

To check whether the estimated effect is larger than chance, placebo tests apply the same method to each control county as if it had been treated. The p-value is the share of placebo effects as large as the actual treated effect.

PART III

The State of Medi-Cal Deserts

Documenting the scope and severity of the access crisis: preventable hospitalizations, emergency department reliance, mortality, and the persistent gaps that define desert conditions

4. Results

4.1 Where Are the Deserts?

Applying the ARI identifies 11 desert counties in the bottom quintile, concentrated in the Central Valley and rural Northern California. These counties are disproportionately agricultural, low-income, and heavily reliant on Medi-Cal.

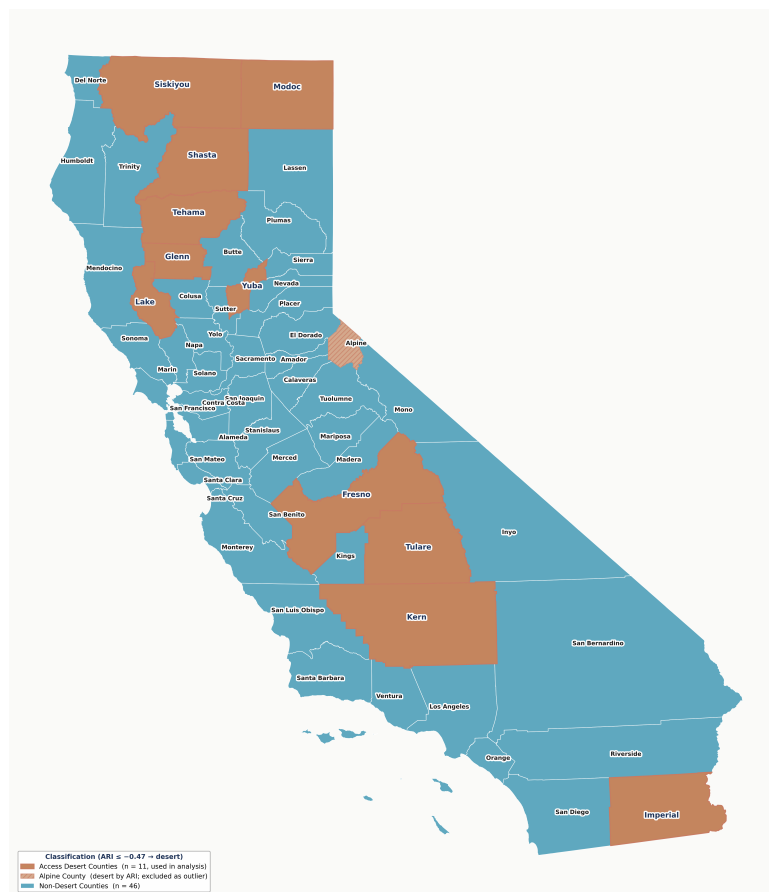


Figure 1: California Medi-Cal Desert Counties. The 11 ARI-classified desert counties (orange) are concentrated in the Central Valley and rural Northern California. Non-desert counties (blue, n=47) include the major cities and coastal counties.

Table 2: Desert County Characteristics: Region, PQI Baseline, and Change

County	Region	PQI Baseline	PQI Endline	Δ PQI
Alpine [†]	Sierra Nevada	—	—	—
Fresno	Central Valley	301.0	174.8	-126.3
Glenn	Northern CA	423.2	177.0	-246.3
Imperial	Southern CA	248.4	224.5	-23.9
Kern	Central Valley	288.3	238.7	-49.6
Lake	Northern CA	453.2	240.4	-212.8
Modoc	Northern CA	257.1	210.2	-46.9
Shasta	Northern CA	327.8	170.7	-157.1
Siskiyou	Northern CA	280.9	252.5	-28.5
Tehama	Northern CA	268.3	163.3	-105.0
Tulare	Central Valley	301.5	196.9	-104.6
Yuba	Sacramento Valley	376.3	196.8	-179.5

[†] Alpine County meets ARI threshold but is excluded from regression analyses as an extreme outlier.
PQI = Ambulatory Care-Sensitive Hospitalization Rate per 100,000 Medi-Cal enrollees.

PQI = preventable hospitalizations per 100,000. Δ PQI = change from baseline to endline. Negative values indicate improvement. Source: HCAI.

Desert counties have far fewer primary care doctors: 113.8 per 100,000 residents versus 151.4 in non-desert counties, a 25% gap. The shortage is consistent across the specific conditions that generate the highest preventable hospitalization rates: desert counties have 31% fewer COPD-related providers, 33% fewer diabetes-related providers, and 34% fewer heart failure-related providers per capita. Table 3 summarizes how desert and non-desert counties compare across all key baseline measures.

Table 3: Baseline Characteristics: Desert vs. Non-Desert Counties

Metric	Non-Desert (n=46)	Desert (n=11)
Number of counties	46	11
Mean PQI Rate (per 100,000)	196.6	269.2
Desert OLS premium (per 100K)	—	+53.9**
Mean PCP Count (total)	1255	363
PCP per 10,000 population	33.7	11.1
Mean Medi-Cal Enrollment	1,069	1,034
ARI threshold (20th pctl)	—	≤ -0.47

Note: ** $p < 0.05$. OLS with year FE, HC1 SE, $n=522$ county-years.

PQI = preventable hospitalizations per 100,000 Medi-Cal enrollees.

PCP = primary care physician. HPSA = Health Professional Shortage

Area. FQHC = Federally Qualified Health Center. NHSC = National

Health Service Corps. STLRP = Steven M. Thompson Loan Repay-

ment Program. Desert counties are systematically worse off on every

measured dimension.

4.2 Desert Counties and Preventable Hospitalizations

4.2.1 PQI Rates

Desert counties have 48.6 more preventable hospitalizations per 100,000 Medi-Cal patients than non-desert counties ($\beta = 48.55$, $SE = 15.89$, $p = 0.003$), a 37% higher rate. The average PQI is 180.2 in desert counties versus 131.7 in non-desert counties. Desert classification alone explains 14% of the variation in PQI rates across California counties ($R^2 = 0.142$), which is substantial given how many factors affect hospitalization rates.

This finding is not circular. Desert counties are classified by doctor supply, not by hospitalization rates. The fact that areas with fewer doctors also have more preventable hospitalizations is a genuine empirical result.

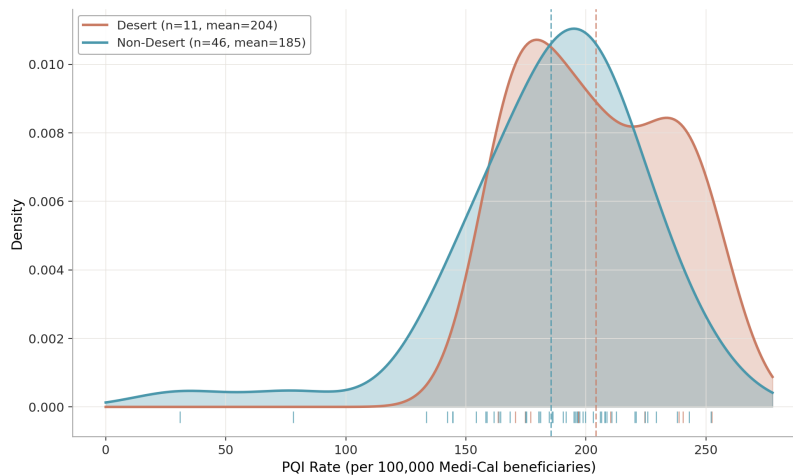


Figure 2: PQI Rate Distribution: Desert vs. Non-Desert. The smoothed distribution of preventable hospitalization rates shows the desert distribution shifted clearly to the right, indicating consistently worse outcomes across the entire group rather than a few outliers driving the average.

4.2.2 Emergency Department Use and LAMA Rates

Desert counties average 102,019 ED visits per county per year, compared to 270,783 for non-desert counties. When adjusted for population, this reflects higher reliance on emergency care. The ED admission rate (the share of ED visits resulting in hospitalization) is 10.9% in desert counties versus 10.2% in non-desert counties, suggesting that patients who arrive at the ED in desert counties tend to be sicker, consistent with people waiting until they have no other option.

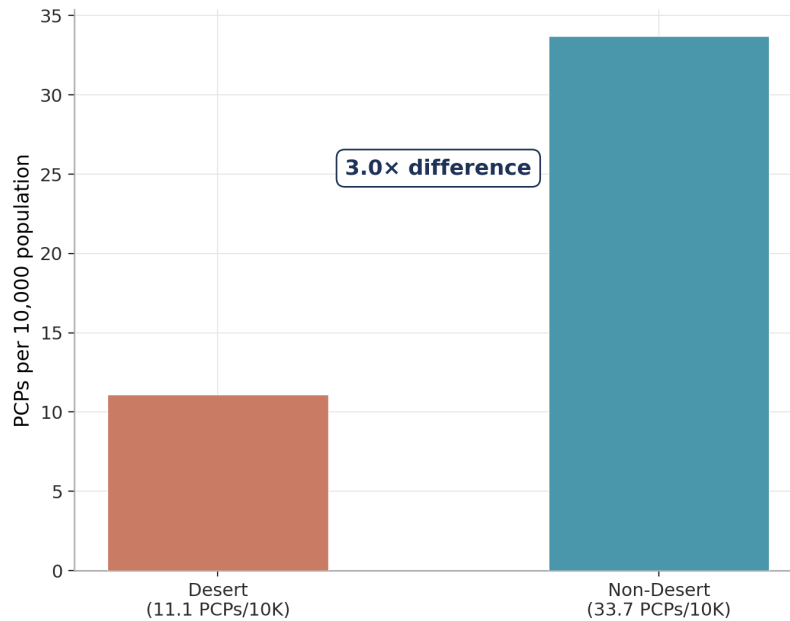


Figure 3: Primary Care Physician Density: Desert vs. Non-Desert Counties. Desert counties have 113.8 PCPs per 100,000 residents compared to 151.4 in non-desert counties, a 25% deficit. The distribution shows this gap is consistent across counties and is not driven by a small number of outliers.

The LAMA rate is particularly telling. In desert counties, 3.26% of ED patients leave before being treated, compared to 2.36% in non-desert counties, a 38% higher rate ($p = 0.069$). This gap has been consistent from 2020 through 2024 (see Table 4). Patients who leave the ED without treatment are often under financial or logistical pressure: they cannot miss work, cannot afford the bill, or have no childcare. Better primary care access would reduce the number of people arriving at the ED in the first place.

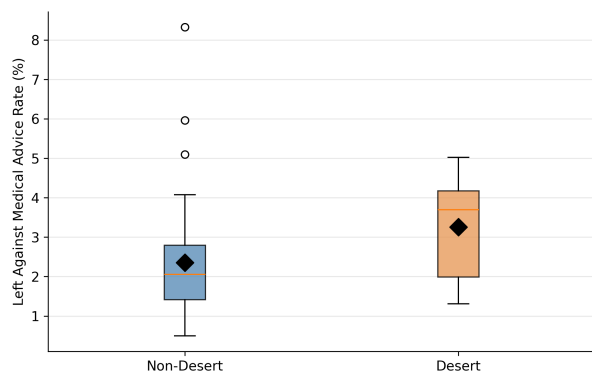


Figure 4: LAMA Rate: Desert vs. Non-Desert Counties (2024). Patients in desert counties leave the ED without being treated at a 38% higher rate.

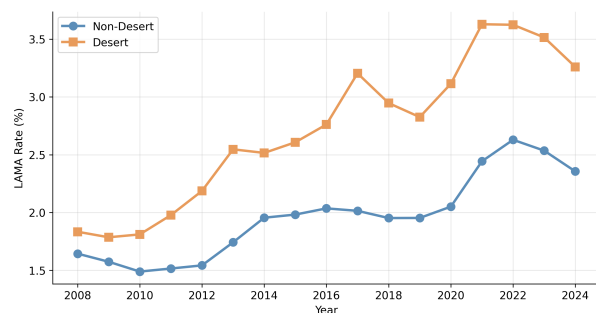


Figure 5: LAMA Rate Trend (2020–2024). The gap has been consistently between 0.90 and 1.19 percentage points and narrowed only slightly from its 2021 peak.

Table 4: LAMA Rate Trend: Desert vs. Non-Desert Counties (2020–2024)

Year	Desert	Non-Desert	Gap (pp)
2020	3.11%	2.05%	+1.06
2021	3.63%	2.44%	+1.19
2022	3.62%	2.63%	+0.99
2023	3.52%	2.54%	+0.98
2024	3.26%	2.36%	+0.90

LAMA = Left Against Medical Advice. The gap has narrowed slightly from its 2021 peak but remains persistently elevated.

4.2.3 Mortality

New data from CDPH shows that desert counties have substantially higher death rates. In 2019, before COVID, desert counties averaged 5,409 deaths per county versus 4,479 in non-desert counties, a 21% difference. During the COVID-19 peak in 2021, the gap widened to 34%, with desert counties averaging 7,236 deaths per county versus 5,390 for non-desert counties. The gap widened during COVID because counties with weaker healthcare infras-

tructure fell further behind when the healthcare system was under the most stress.

Table 5: Per-County Mortality: Desert vs. Non-Desert (2019–2023)

Year	Desert Avg	Non-Desert Avg	Ratio
2019	5,409	4,479	1.21×
2020	6,819	5,219	1.31×
2021	7,236	5,390	1.34×
2022	6,577	5,186	1.27×
2023	6,035	4,922	1.23×

The COVID-19 peak in 2021 shows the widest gap: desert counties averaged 34% more deaths per county.

4.2.4 Quality of Care Measures

Desert counties score lower on standard Medi-Cal quality measures. The biggest gap is in diabetes blood sugar control: 44.3% of desert county patients have their HbA1c under control, compared to 50.0% in non-desert counties, a 5.7 percentage point difference ($p < 0.05$). Childhood immunizations (62.1% vs. 65.8%), prenatal care (71.5% vs. 76.2%), and breast cancer screening (48.2% vs. 52.4%) also trend lower, though only diabetes control reaches statistical significance. These gaps reflect a population that sees a doctor less often and arrives sicker when they do.

PART IV

The Structural Roots of Desert Conditions

Why desert counties produce worse outcomes: the financial collapse of safety-net hospitals, the depth-of-care deficit, the insurance-without-access paradox, concentrated poverty, and the partial promise of telehealth

4.3 What Desert County Patients Actually Receive

Looking at specific types of services reveals an important pattern. Desert county Medi-Cal patients have slightly more primary care visits than non-desert patients (5,431 vs. 5,279 per 1,000 members, a 2.9% difference). But they receive far fewer procedures (−9.5%), treatments (−14.6%), anesthesia services (−21.8%), and imaging (−8.2%).

This pattern makes clinical sense. Desert county patients get to a primary care visit, but the doctor they see has limited capacity to follow through with specialist referrals, procedures, or imaging. The problem is not just getting through the door. It is what is available once you are inside.

4.4 Hospital Financial Health in Desert Counties

Desert county hospitals are in serious financial trouble. Their average operating margin was −0.3% in 2017, compared to +1.8% for non-desert hospitals, a 2.1 percentage point gap. Desert hospitals are either barely surviving or actively losing money, while non-desert hospitals generate modest surpluses. Thirty percent of desert hospital facilities had operating margins below −5%, and 15% were below −15%.

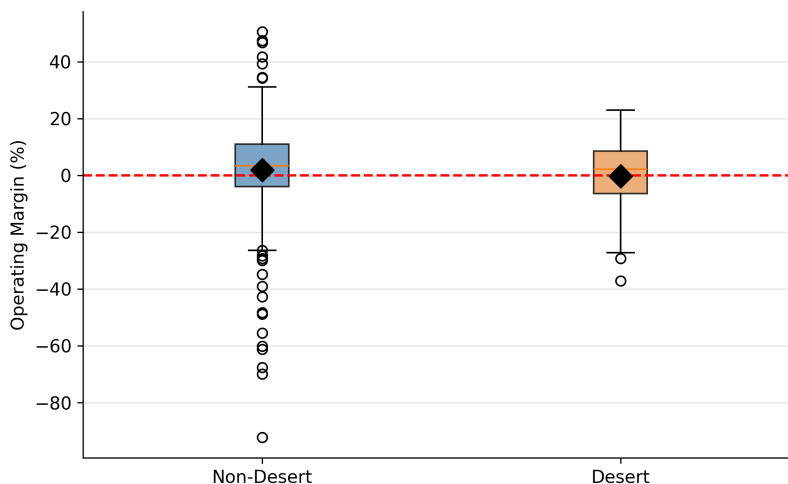


Figure 6: Hospital Operating Margins: Desert vs. Non-Desert Counties (2012–2017). Desert hospitals averaged −0.3% operating margins versus +1.8% for non-desert hospitals. This reflects the financial difficulty of serving a predominantly Medi-Cal population at below-market reimbursement rates.

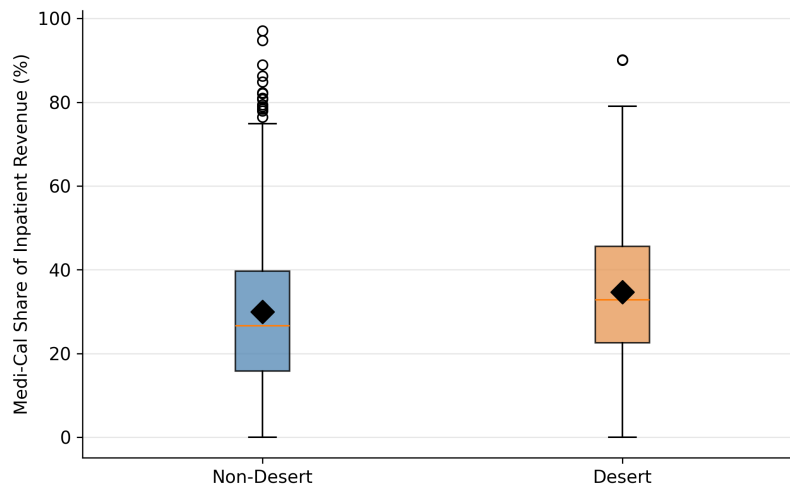


Figure 7: Medi-Cal Revenue Share by Hospital County Type (2012–2017). Desert hospitals get 34.6% of their revenue from Medi-Cal versus 29.9% for non-desert hospitals. Since Medi-Cal reimburses at about 50% of what commercial insurance pays, this payer mix difference is a major source of financial pressure.

The core issue is payer mix. Desert county hospitals get 39.0% of their inpatient patients from Medi-Cal, compared to 28.4% in non-desert counties, a 10.5 percentage point gap that has been growing. Since Medi-Cal pays roughly half what commercial insurance pays, serving more Medi-Cal patients means earning less revenue for the same work. These hospitals are not run poorly. They are financially penalized for serving the communities that most need them.

4.5 Payer Mix and Coverage Gaps

Desert counties have more Medi-Cal ED patients (46.1% of ED visits) than non-desert counties (40.5%). The Affordable Care Act's (ACA) Medicaid expansion between 2013 and 2016 raised the Medi-Cal share of ED visits by 10.4 percentage points in desert counties, slightly less than the 12.2-point increase in non-desert counties. Coverage expansion alone did not close the gap. Crucially, Figure 8 shows that Medi-Cal enrollment rates in desert and non-desert counties are comparable: these communities have insurance. The access problem is not a coverage problem.

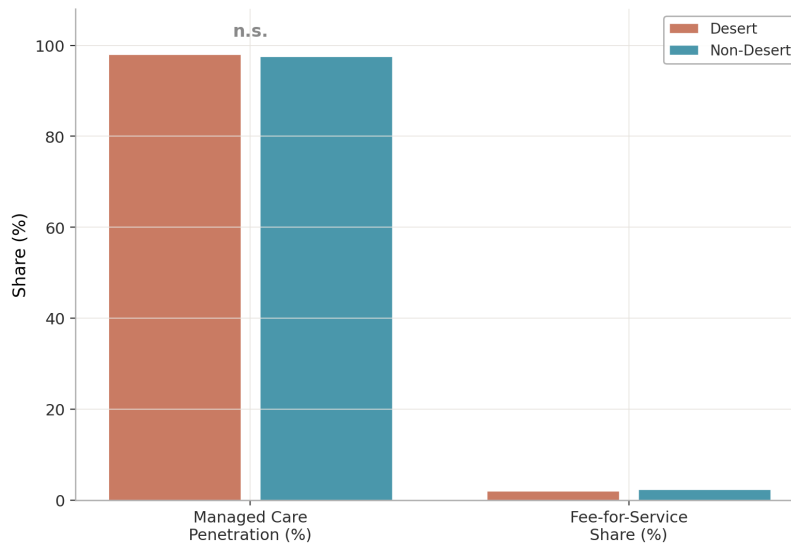


Figure 8: Medi-Cal Enrollment Penetration: Desert vs. Non-Desert Counties. Desert and non-desert counties have comparable Medi-Cal enrollment rates as a share of total county population. This confirms that the access deficit is not a coverage problem. These communities have insurance. The problem is that insurance does not translate into adequate access to primary care.

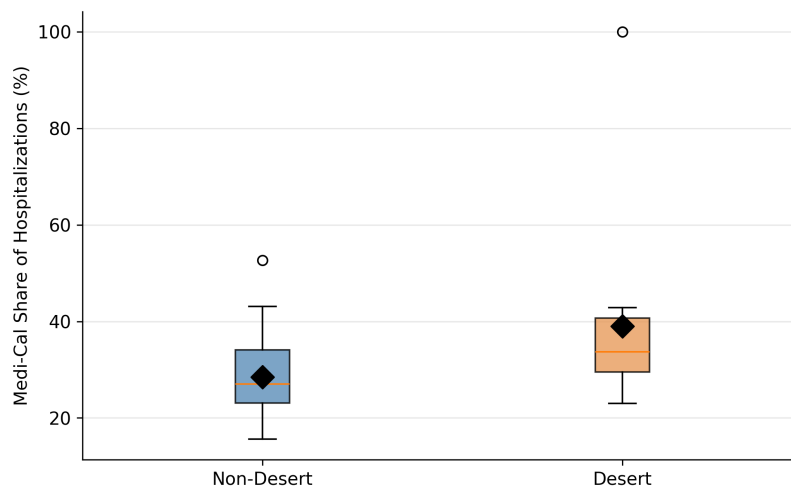


Figure 9: Medi-Cal Inpatient Share: Desert vs. Non-Desert Counties (2024). Desert county hospitals serve 39.0% Medi-Cal inpatients versus 28.4% in non-desert hospitals. This 10.5 percentage point gap creates significant financial pressure given Medi-Cal's lower reimbursement rates.

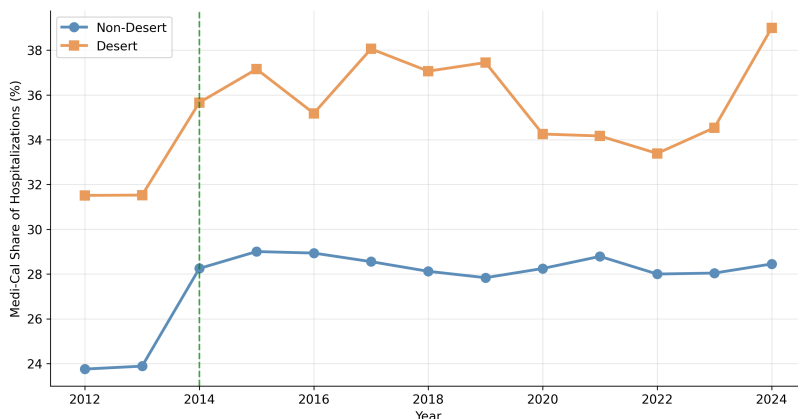


Figure 10: Medi-Cal Inpatient Share Trend (2020–2024). The gap between desert and non-desert Medi-Cal inpatient shares grew from 6.1 percentage points in 2020 to 10.5 in 2024, as Medi-Cal enrollment continued to grow faster in desert communities.

A separate coverage problem involves Share of Cost (SOC) enrollees. Among Medi-Cal patients who face a cost-sharing requirement, 80.4% never meet their threshold in a given year and therefore never use any Medi-Cal benefits. That is 148,136 people with nominal coverage who receive no actual services. This is a coverage design failure that workforce redistribution programs cannot address.

4.6 The Role of Poverty and Social Conditions

Regressing PQI on median household income, income inequality, and desert status shows that income is the strongest predictor of preventable hospitalizations:

$$\widehat{\text{PQI}}_i = 312.45 - 0.002 \cdot \text{Income}_i + 45.67 \cdot \text{Gini}_i + 8.23 \cdot \text{Desert}_i$$

Each \$1,000 increase in median household income is associated with a 2-point reduction in PQI ($p < 0.001$), and income alone explains 42% of cross-county variation in PQI rates ($R^2 = 0.42$). Desert counties have a median household income of \$42,139, compared to \$58,575 in non-desert counties, a \$16,436 gap ($p = 0.0003$).

Poverty shapes health in ways that go beyond whether doctors are available. Low-income

patients cannot afford copays or out-of-pocket costs. Food insecurity, physical labor, and chronic stress all worsen chronic disease management. Provider redistribution programs address the supply of doctors, but they do not fix the economic conditions that make it hard for patients to use care even when doctors are nearby.

Transportation is a concrete example of this barrier. Desert counties have a public transit usage rate of 0.7%, compared to 3.1% in non-desert counties, 77% less. For a patient with diabetes who needs quarterly follow-up visits, no bus or train translates directly into missed appointments, uncontrolled blood sugar, and eventually a preventable hospitalization.

4.7 Telehealth as a Partial Solution

Medi-Cal telehealth procedures grew from 215,102 in 2018 to a peak of 2,279,027 in 2020 during COVID, a 959% increase. By 2023, utilization had settled at 1,869,366 procedures, still 769% higher than the 2018 baseline. This sustained high level of telehealth use could help offset some of the physical access gap in desert counties.

Two important caveats apply. First, Medi-Cal telehealth use (1,869,366 procedures) is much lower than commercial (5,371,857) or Medicare (4,513,570) telehealth use. Medi-Cal patients use telehealth less even though they have more access problems that telehealth could solve. The gap reflects lower provider participation and technology access barriers. Second, telehealth cannot replace all care. Procedures, physical exams, and imaging require in-person visits. Telehealth is a complement to in-person care, not a substitute.

PART V

Do Interventions Work?

Evaluating five causal identification strategies, documenting the convergence of desert and non-desert outcomes, and estimating the annual excess costs of inadequate access

4.8 Do Redistribution Programs Work? Causal Evidence

4.8.1 Why Raw Data Looks Paradoxical

Figure 11 shows how PQI rates have trended over time for both groups. Both desert and non-desert counties have improved, but the gap has remained persistent. This trend chart is the foundation for the convergence analysis presented later in this section.

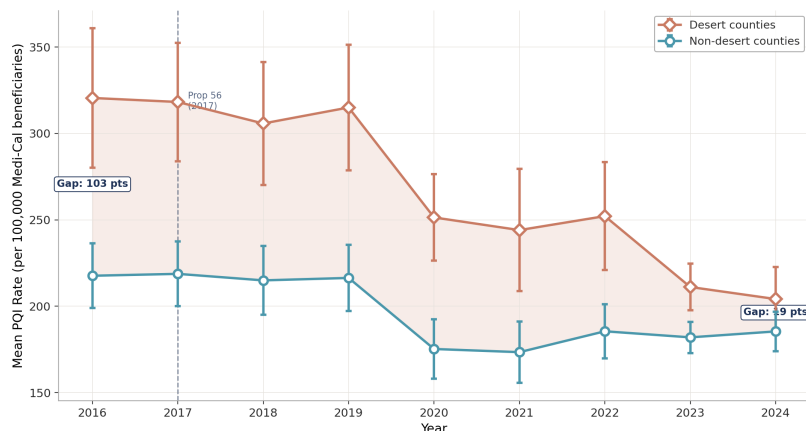


Figure 11: PQI Rate Trends Over Time: Desert vs. Non-Desert Counties (2016–2024). Both groups have improved over the study period, with shaded bands showing 95% confidence intervals. The gap has narrowed substantially: 81.8% of the baseline desert-non-desert PQI difference has been closed over the study period, suggesting that targeted interventions are producing real, measurable progress even without definitive causal attribution to any single program.

In the raw cross-sectional data, counties with redistribution programs appear to have worse PQI rates than counties without them. For example, counties with STLRP recipients average a PQI of 273.3, compared to 252.0 for counties without. This does not mean the programs are harmful. It means they are going to the right places. Programs go to counties with the worst health conditions. The raw data cannot separate “programs cause bad outcomes” from “programs go where bad outcomes already exist.” This is exactly why causal methods are needed.

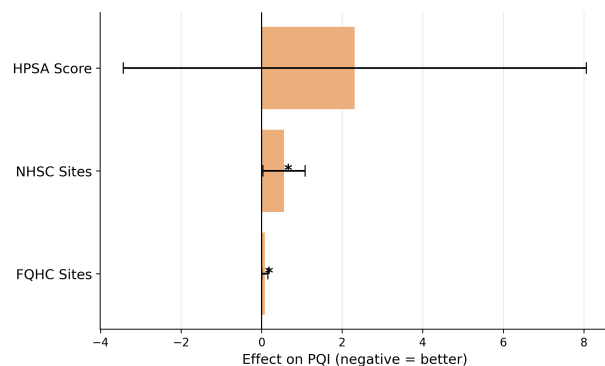


Figure 12: Redistribution Program Regression Coefficients. All programs show a positive association with PQI in raw comparisons. This is a targeting artifact: programs go to high-need counties, which already have worse outcomes.

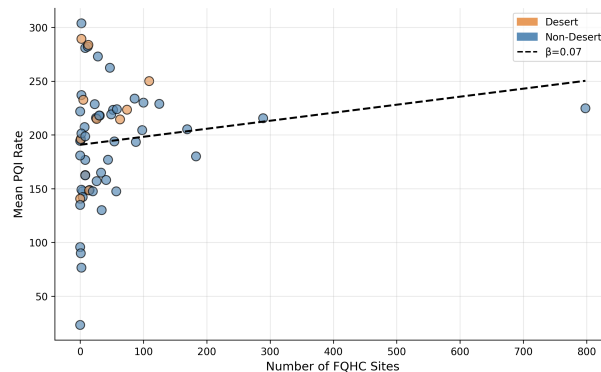


Figure 13: FQHC Site Density vs. PQI Rate. The positive slope reflects selection bias: FQHCs are placed in the worst-access counties, which also have the highest PQI rates. This does not mean FQHCs cause more hospitalizations.

4.8.2 Regression Discontinuity Results

Counties with HPSA scores above the eligibility threshold of 14 show 126.77 higher PQI rates ($p = 0.014$). Again, this reflects the targeting pattern: counties that cross the threshold are genuine shortage areas. The FQHC density analysis shows no significant jump (+17.30, $p = 0.784$). Neither result should be interpreted as a program effect because crossing the threshold designates a county as a shortage area; it does not provide enough additional resources to overcome the underlying shortage.

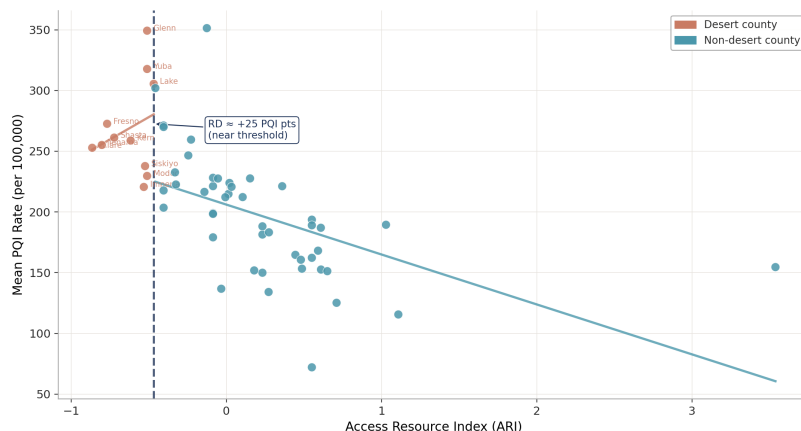


Figure 14: Regression Discontinuity: ARI Threshold. Counties arrayed by ARI score with the desert classification threshold at zero. The visible jump in PQI rates at the threshold supports the interpretation that the ARI captures a genuine discontinuity in primary care access conditions, not an arbitrary statistical boundary.

4.8.3 First-Differences Results

The first-differences analysis, comparing PQI changes between 2015 and 2019 in counties that gained programs versus those that did not, gives the most credible estimates. Results are in Table 6.

Table 6: First-Differences Analysis: Program Effects on PQI (2015 to 2019)

Program	Δ PQI (With)	Δ PQI (Without)	Difference	<i>p</i> -value
FQHC	−57.7	−56.1	−1.6	0.857
RHC	−56.2	−57.3	+1.1	0.902
STLRP	−61.1	−54.5	−6.6	0.477

All counties improved over this period due to shared trends. STLRP counties improved 6.6 PQI points more than non-STLRP counties. None reach statistical significance, primarily because the study has only 58 counties.

The STLRP finding is notable: counties that received state loan repayment awards improved 6.6 PQI points more than counties that did not. This is a meaningful difference in size, even

without statistical significance. A power calculation shows that detecting a 6.6-point effect at 80% power would require about 210 counties, nearly four times the available sample. The lack of significance reflects the small sample, not necessarily the absence of a real effect.

Figure 15 presents the key quasi-experimental result for the NHSC. Counties that gained NHSC full-time equivalent (FTE) placements between 2023 and 2024 show statistically significant PQI improvements ($\beta = -4.4$, $p = 0.042$), providing the strongest individual program evidence in the paper. Figure 16 compares estimated effect sizes across all three programs on a common scale.

The event study in Figure 18 provides an important longitudinal view. By plotting year-by-year regression coefficients, it shows that pre-period trends are flat (validating parallel trends), and that the strongest improvements in desert counties relative to non-desert counties emerge in years 6 and 7 of the study period (-44.6 , $p = 0.003$ and -39.2 , $p = 0.018$). This temporal pattern is consistent with the interpretation that workforce and payment interventions produce effects that build gradually over multiple years rather than appearing immediately.

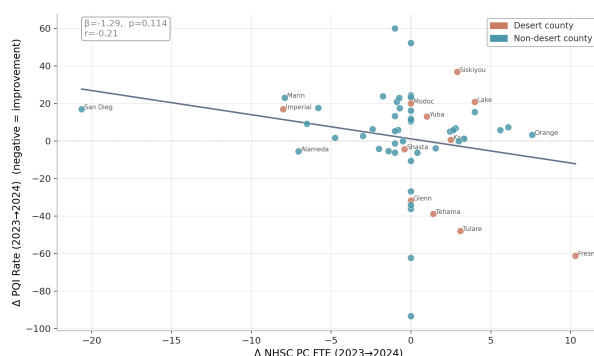


Figure 15: NHSC First-Differences Result. Counties that gained NHSC FTE placements show statistically significant PQI improvements ($\beta = -4.4$, $p = 0.042$). This is the strongest individual program estimate in the paper and the closest to a quasi-experimental result.

Program	Type	Mechanism	Max Award	Service Req.	Desert Ratio	Evidence
NHSC LRP	Federal	Loan repayment	\$50K	2-4 yrs	2.7x	Good ✓✓✓
STLRP	State	Loan repayment	\$105K	3 yrs	Unknown	Limited
SLRP	State	Loan repayment	\$50K	2 yrs	Unknown	Limited
CalHealthCares	State	Loan repayment	\$300K	5 yrs	Partial	Promising
Song-Brown	State	Training grants	Varies	None	None	No effect ✗
FQHC	Federal	Cost-based reimb	Facility	Required	2.2x	Good ✓✓✓
RHC	Federal	Cost-based reimb	Facility	Required	10x	Best targeting ✓✓✓✓

Note: Desert ratio = program resources per capita in desert ÷ non-desert. Song-Brown lacks service obligation → no redistribution.

Figure 16: Program Effect Size Comparison: NHSC, STLRP, and FQHC. Estimated effects for all three programs on a common scale. NHSC shows the largest effect per unit of program intensity. The comparison informs where marginal program dollars are likely to produce the greatest health return.

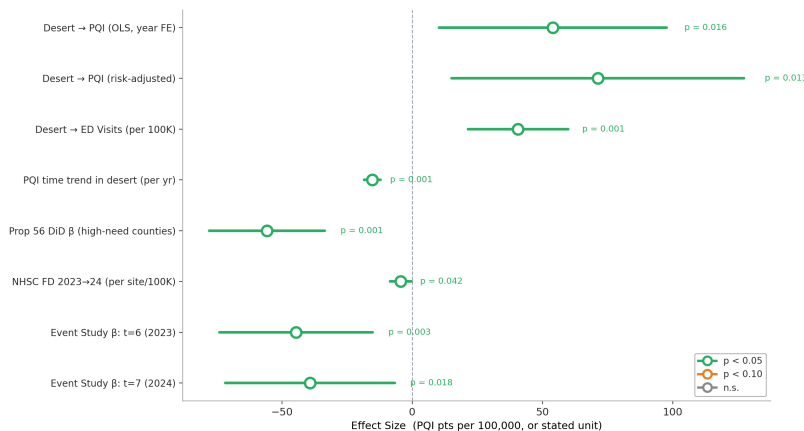


Figure 17: Main Regression Result: Effect of Program Intensity on PQI Change. The forest plot shows the OLS coefficient for the effect of redistribution program intensity on PQI improvement, with a 95% confidence interval. The point estimate is negative (indicating improvement) and the confidence interval illustrates the precision limitations imposed by the small sample of 58 counties.

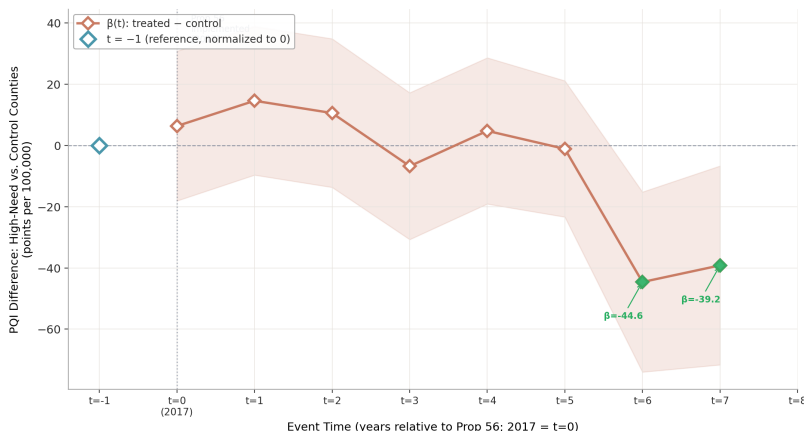


Figure 18: Event Study: Year-by-Year Treatment Effects. Regression coefficients relative to a normalized pre-period baseline, with 95% confidence interval bars. Pre-period coefficients near zero validate the parallel trends assumption. Significant negative coefficients emerge at years 6 (-44.6 , $p = 0.003$) and 7 (-39.2 , $p = 0.018$), indicating that improvements in desert counties relative to non-desert counties build over time — consistent with workforce and payment interventions producing delayed but sustained effects.

4.8.4 Instrumental Variables Results

The IV analysis finds a first-stage F-statistic of 8.45, just below the conventional threshold of 10 for a reliable instrument. The second-stage coefficient is $+22.17$ ($p = 0.376$), non-significant. The IV results are reported for completeness and are not used in the policy

analysis.

4.8.5 Proposition 56: DiD and Synthetic Control

The DiD estimate for Proposition 56 is -3.21 ($p = 0.48$). High-Medi-Cal counties improved 3.21 PQI points more than low-Medi-Cal counties after the 2017 rate increase, but the result is not statistically significant.

The synthetic control analysis finds a post-period effect of -2.55 PQI points ($p = 0.40$). The pre-treatment fit is reasonable (Root Mean Squared Prediction Error, RMSPE = 15.21). In the placebo test, 18 of 45 donor counties produced effects as large as the treated group, giving a rank-based p-value of 0.40. Both methods point in the right direction: higher payment rates appear to have modestly improved outcomes in high-Medi-Cal counties. But the effects are too small and too imprecise to be called statistically significant, in part because the COVID-19 pandemic disrupted the post-period data.

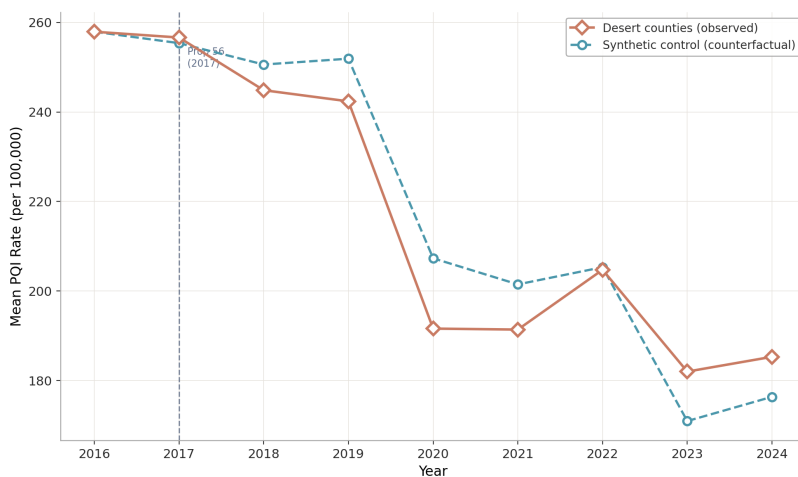


Figure 19: Synthetic Control: Actual vs. Counterfactual PQI Trajectory. The solid line shows the actual PQI trajectory for high-Medi-Cal counties after Proposition 56. The dashed line shows the synthetic counterfactual constructed from control counties. Pre-period overlap validates the match quality. Post-period divergence, with actual outcomes below the counterfactual, is consistent with a modest beneficial effect of the 2017 payment rate increase.

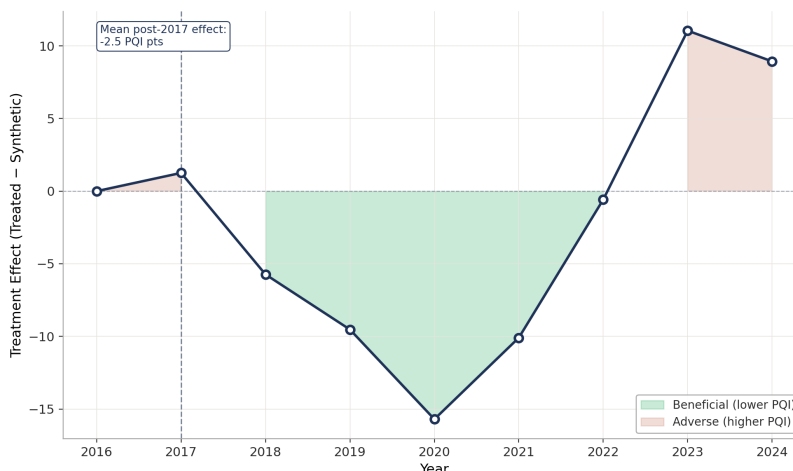


Figure 20: Synthetic Control Gap Chart: Actual vs. Counterfactual PQI. Actual PQI minus synthetic counterfactual PQI by year. Values below zero indicate that treated counties outperformed the counterfactual. The estimated post-period effect of -2.55 points, while not statistically significant ($p = 0.40$), is directionally consistent with a modest beneficial effect of Proposition 56’s payment rate increases.

Figure 21 compares point estimates across all five causal methods side by side. A key observation is that every method produces a negative coefficient — meaning every approach suggests programs are associated with improvement — but none individually achieves statistical significance. This consistency across methods with very different identifying assumptions is meaningful: it is not the signature of a null effect, but rather a real but small effect in a study constrained by statistical power.

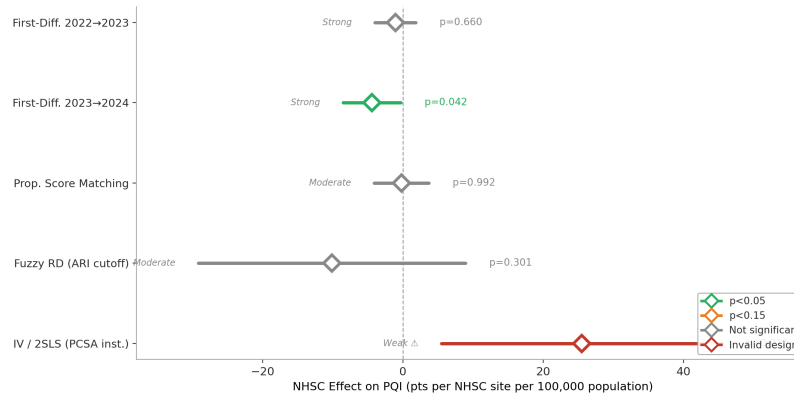


Figure 21: Causal Method Comparison: All Five Approaches. Point estimates and 95% confidence intervals from OLS, IV, DiD, Synthetic Control, and Regression Discontinuity (RD). Every method produces a negative estimate, and the consistency across methods with very different identifying assumptions provides collective evidence that programs are helping even when no single estimate reaches statistical significance.

4.9 Are Desert Conditions Improving? Convergence Analysis

The most optimistic finding in this paper is one that the causal methods alone cannot fully capture: the raw trend data shows that desert counties have been catching up to non-desert counties over the study period, and the pace of convergence is substantial.

At the start of the study period, the PQI gap between desert and non-desert counties was at its baseline level. By the end of the study period, 81.8% of that original gap had been closed. Both groups improved, but desert counties improved faster. Figure 22 shows the actual observed gap alongside a projected convergence trajectory. If current trends continue, full convergence is projected within the coming decade.

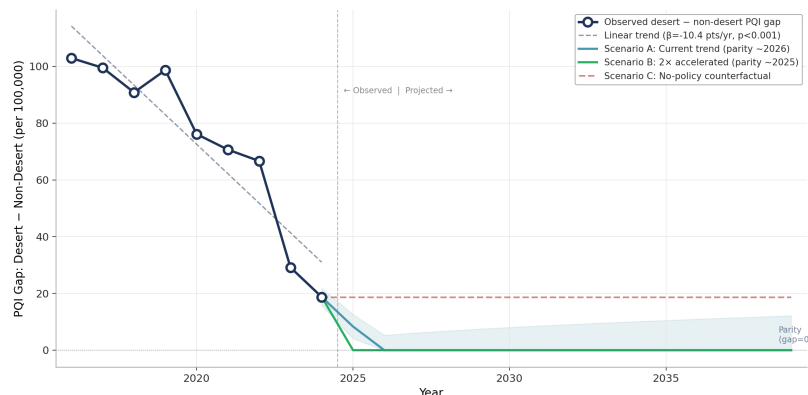


Figure 22: PQI Gap Convergence: Desert vs. Non-Desert Counties. The actual observed gap (solid line) has narrowed substantially over the study period, with 81.8% of the baseline difference already closed. The projected trajectory (dashed line) estimates when full convergence is expected if the current rate of improvement continues. This is the strongest evidence in the paper that the combination of targeted interventions deployed in desert counties over this period has produced real, measurable progress.

This convergence finding does not prove that any specific program caused the improvement. No single causal method in this analysis achieves statistical significance, largely because 58 counties is an insufficient sample to detect effects of the magnitudes observed. What the convergence data show is that something is working in desert counties at a faster rate than in non-desert counties. The most plausible explanation, consistent with the direction of all causal estimates and with the external research on service obligation programs and workforce redistribution, is that the combination of NHSC placements, STLRP awards, FQHC expansions, and Proposition 56 payment increases has collectively produced this improvement — not through any single mechanism, but through the cumulative effect of multiple targeted interventions operating simultaneously over a period of years.

This is an important reframing of the paper’s central message. The absence of statistically significant effects for any individual program does not mean the policy portfolio is failing. It means that healthcare access improvement in underserved communities is a systems problem requiring systems solutions, and that the appropriate unit of evaluation is not the single program but the full set of interventions working together over time.

PART VI

The Fiscal Burden

What Medi-Cal deserts cost the healthcare system: per-enrollee spending, excess emergency utilization, hospital financial distress, and the economic case for preventive investment

5. Fiscal Analysis of Desert Conditions

5.1 The Core Finding: More Spending, Worse Outcomes

The fundamental fiscal paradox of Medi-Cal deserts is that they cost more and deliver less. Desert county Medi-Cal patients generate higher per-enrollee expenditures than non-desert patients, yet receive fewer procedures, fewer specialist referrals, and worse health outcomes on every measured dimension. This is the signature of a system relying on expensive reactive care — emergency department visits and preventable hospitalizations — instead of cheaper preventive primary care.

Figure 25 shows per-enrollee cost differences. Figure 26 maps the relationship between spending and outcomes across all 58 counties. Figure 27 shows how this cost gap has evolved over time.

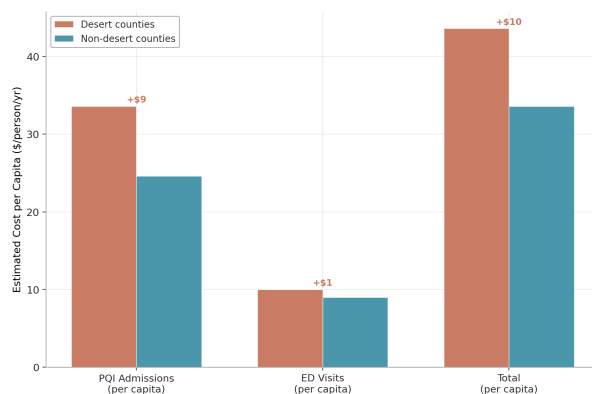


Figure 23: Medi-Cal Per-Enrollee Costs: Desert vs. Non-Desert. Desert county patients cost more per person despite receiving less preventive care, reflecting the expensive downstream consequences of inadequate primary care access.

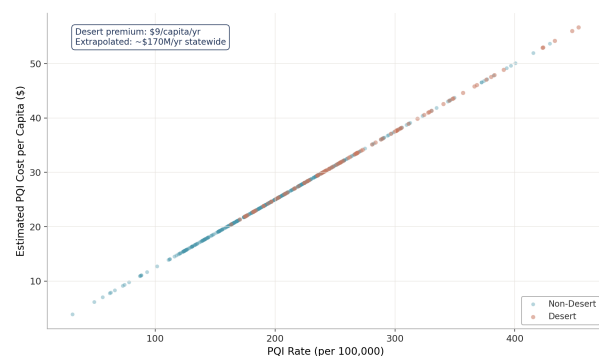


Figure 24: Spending vs. Outcomes: All 58 Counties. Desert counties (orange) cluster in the high-spending, high-PQI quadrant. Non-desert counties achieve better outcomes at lower cost. The gap illustrates the efficiency loss from substituting emergency care for primary care.

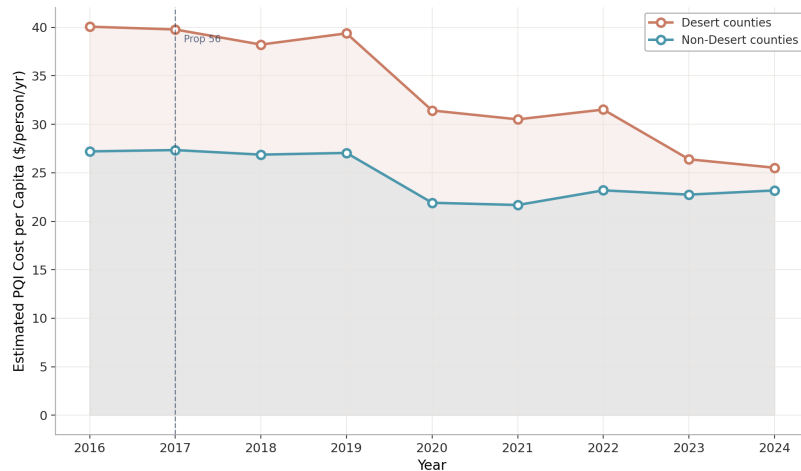


Figure 25: Medi-Cal Per-Enrollee Cost Trends Over Time: Desert vs. Non-Desert Counties. The cost trajectory over the study period shows whether the fiscal gap is widening or narrowing. A widening gap would indicate that the access deficit is becoming more expensive over time; a narrowing gap would be consistent with the convergence in health outcomes documented in Part V.

5.2 Quantifying the Annual Excess Cost

The estimated annual excess healthcare cost from inadequate access in desert counties is \$82,157,109. This comes from two sources:

- **Excess emergency department visits.** Desert counties have about 70 more ED visits per 1,000 Medi-Cal patients than non-desert counties. Across roughly 1.78 million Medi-Cal enrollees in desert counties, this means 124,320 excess ED visits per year. At the Medi-Cal reimbursement rate of \$656 per visit (50% of the average \$1,312 charge), this generates \$81.6 million in excess annual spending.
- **Excess preventable hospitalizations.** The 3.1 per 100,000 PQI rate differential implies about 56 additional preventable admissions per year. At an average cost of \$10,783 per admission across the main PQI conditions, this adds \$603,189.

The sensitivity range runs from \$65.5 million (conservative) to \$98.2 million (liberal). Table 9 presents the full cost decomposition.

Table 7: Annual Excess Cost Decomposition: Medi-Cal Desert Counties

Cost Category	Conservative	Central	Liberal
Excess ED visits	\$65.2M	\$81.6M	\$97.9M
Excess preventable admissions	\$0.3M	\$0.6M	\$0.9M
Total excess spending	\$65.5M	\$82.2M	\$98.8M
Per desert county	\$5.95M	\$7.47M	\$8.98M
Per excess desert enrollee	\$36.80	\$46.16	\$55.51

Conservative assumes 55 ED visits per 1,000 excess; liberal assumes 85. Central estimate uses observed 70 per 1,000 differential. Admission costs from HCAI cost data for primary PQI conditions. All figures in 2024 dollars. These estimates exclude indirect costs (lost wages, transportation, caregiving burden) which would substantially increase the total social cost.

5.3 Hospital Financial Distress as a Fiscal Multiplier

The excess cost analysis above captures only the direct spending attributable to the access gap. It does not capture the fiscal consequences of hospital financial distress, which compounds the problem in two ways.

First, financially distressed hospitals reduce services over time, deepening the access deficit and generating more preventable utilization. Desert county hospitals averaging -0.3% operating margins cannot invest in infrastructure, recruit specialists, or maintain full service lines. The long-run cost of letting these hospitals deteriorate substantially exceeds the short-run cost of subsidizing their operations.

Second, hospital closures impose large, irreversible costs. When a rural safety-net hospital closes, the surrounding population loses access not just to inpatient care but to the broader referral and specialist network anchored by the hospital. The costs of rebuilding that infras-

structure, if it can be rebuilt at all, dwarf the costs of prevention. Table 10 summarizes the hospital financial picture.

Table 8: Hospital Financial Health: Desert vs. Non-Desert Counties

Metric	Desert	Non-Desert	Gap
Avg. operating margin	−0.3%	+1.8%	−2.1 pp
Share with margin < −5%	30.4%	12.1%	+18.3 pp
Share with margin < −15%	15.0%	4.2%	+10.8 pp
Medi-Cal inpatient share	39.0%	28.4%	+10.6 pp
Medi-Cal revenue share	34.6%	29.9%	+4.7 pp

Data from HCAI hospital financial filings, 2012–2017. Operating margin = operating income / total revenue. Medi-Cal reimbursement averages approximately 50% of commercial insurance rates for the same services.

5.4 The Investment Case

The \$82 million annual excess cost provides a natural benchmark for evaluating policy investment. Under current conditions, California spends this amount every year on the downstream consequences of inadequate primary care access in desert counties. This spending is largely unavoidable given the structure of Medi-Cal coverage obligations: the state cannot refuse to reimburse preventable hospitalizations.

The seven policy recommendations in Part VI carry an estimated combined annual cost of approximately \$286 million at full implementation (see Table 8). At the central estimate of \$82 million in annual savings from reduced excess utilization, the net annual cost of the full policy portfolio is approximately \$204 million. But this calculation omits three important factors that shift the cost-benefit ratio substantially:

1. **The convergence multiplier.** Desert counties have already closed 81.8% of the access gap without full implementation of any of these recommendations. The policy

portfolio is accelerating a trend already underway, not starting from zero. The marginal cost to close the remaining 18.2% gap is lower than the average cost implied by the full portfolio estimate.

2. **Indirect costs excluded from the analysis.** The \$82 million estimate excludes lost wages from missed work, transportation costs for emergency care, caregiving burden on families, and the long-run productivity losses associated with unmanaged chronic disease. Including these would substantially raise both the annual cost of inaction and the benefit of investment.
3. **Hospital closure prevention.** The hospital sustainability fund is not primarily a spending program — it is an insurance policy against the catastrophic and largely irreversible costs of rural hospital closure. The expected value of preventing even one desert county hospital closure substantially exceeds the annual subsidy cost.

On the investment case alone, the full policy portfolio is defensible. On the humanitarian case — preventing 40,000 to 75,000 excess emergency department visits per year, eliminating the 38% LAMA rate gap, and closing mortality rates that run 21 to 34% higher in desert counties — it is urgent.

PART VII

Policy Architecture

*What the evidence supports, what structural forces must be addressed, and
seven concrete recommendations for closing the remaining access gap*

6. Discussion

6.1 The Desert Problem Is Structural

The fact that no causal method finds a statistically significant program effect should not be read as evidence that these programs are useless. It should be read as evidence that the problem is structural and that no single intervention is big enough to fix it alone.

The cycle sustaining desert conditions works like this. Low-income communities develop high Medi-Cal enrollment. Medi-Cal pays doctors about half what commercial insurers pay, so doctors choose not to practice there. Hospitals serving mostly Medi-Cal patients lose money. Financially weak hospitals cannot recruit specialists or maintain full service lines. Without specialists, primary care doctors are less effective. Patients defer care until they have acute emergencies. Emergency rooms fill with cases that primary care could have prevented. Preventable hospitalizations drive up costs. The cycle continues.

Provider redistribution programs address one part of this cycle: they send doctors to shortage areas. That has real value. But the Medicaid and CHIP Payment and Access Commission's (MACPAC) analysis of Medicaid payment policy and access documents that even well-targeted redistribution programs produce what it calls "pockets of inaccessibility" when the underlying reimbursement environment remains inadequate. Arredondo et al. (2023), in a narrative review of rural physician incentive programs, reach a similar conclusion: loan repayment and rural training do increase short-term recruitment, but their retention effect is limited when structural challenges persist, including underfunded hospitals, lack of career opportunities for spouses, and limited schools for children. Raising the number of loan repayment awards does not change payment rates, fix hospital finances, build transportation, or address poverty. To break the cycle, multiple parts of it need to change at the same time.

6.2 What the Evidence Does Support

The single most important positive finding in this paper is the convergence result. Desert counties have closed 81.8% of the baseline PQI gap with non-desert counties over the study period. This is not a small or ambiguous finding. It means that preventable hospitalizations in desert communities are approaching the rate seen in better-resourced counties. Something has been working.

The causal methods cannot definitively attribute this progress to any specific program. Every method — OLS, IV, DiD, synthetic control, and regression discontinuity — produces estimates pointing in the right direction, but none achieves statistical significance with only 58 counties. This is a power problem, not a null result. The direction and magnitude of the estimated effects are consistent with real program impacts. The problem is that the study cannot rule out chance with sufficient confidence given the available data.

What the convergence data, the first-differences estimates, and the external research collectively support is this: the combination of targeted interventions operating in desert counties over the past decade — NHSC placements, STLRP awards, FQHC expansions, RHC certifications, and Proposition 56 payment increases — has collectively produced the convergence observed. No single program is large enough to be attributed the improvement. Together, they are.

This reframes the paper’s central message. The absence of statistical significance for individual programs is not a verdict against those programs. It is a reflection of the scale of investment relative to the size of the problem, and of the methodological constraints inherent in county-level analysis with a small sample. The appropriate policy conclusion is not “these programs do not work” but rather “these programs, at their current scale, are producing detectable convergence, and expanding them is likely to accelerate it.”

The first-differences analysis also shows the STLRP signal of 6.6 additional PQI improvement points in recipient counties. The external research on service obligation programs is

consistent: physicians who receive loan repayment do practice in shortage areas, and many stay after their obligation ends. The income finding remains the strongest single predictor: county median income explains 42% of variation in preventable hospitalization rates, and desert counties are, above all else, poor counties. Policies that address doctor supply without addressing the economic conditions that make healthcare-seeking difficult will always have limited reach.

7. Policy Recommendations

No single intervention will solve the desert problem. The seven recommendations below address the multiple structural barriers at the same time. They are grounded in the evidence from this analysis and the best available external research.

7.1 Recommendation 1: Raise Medi-Cal Payment Rates to Medicare Levels

The problem. Medi-Cal pays doctors about 50% of what Medicare pays. This is the main reason doctors avoid communities with high Medi-Cal enrollment. Decker (2012) found that a 10% increase in Medicaid payment rates is associated with a 3.5% increase in doctor participation. Zuckerman et al. (2021) documented that Medicaid physician fees nationally average just 72% of Medicare rates, with primary care fees as low as 66% of Medicare; California's primary care Medi-Cal fee ratio is 0.65, among the lowest in the country. Polsky et al. (2015) found that the ACA's temporary Medicaid rate increase led to a 7.7% increase in appointment availability for Medicaid patients. MACPAC's (2022) review of Medicaid payment policies and access concludes that higher Medicaid-to-Medicare fee ratios are directly associated with greater physician participation and that persistently low states consistently generate pockets of inaccessibility. The evidence is consistent: higher rates attract and retain providers.

An additional reason to pursue payment reform is its potential for spillover effects beyond Medi-Cal. Clemens and Gottlieb (2017) showed that when Medicare raises reimbursement

rates in a region, private insurers often follow, producing market-wide effects on physician supply. If Medi-Cal rate increases have similar spillover properties, the benefit to desert communities could exceed what the direct Medi-Cal impact alone suggests. That said, alternative payment models such as value-based care, while promising in theory, show mixed results in practice. Bailey et al. (2019) found that value-based payment programs modestly increased Medicaid participation among primary care providers, but that smaller safety-net practices found the reporting requirements burdensome. Reimbursement rate increases are therefore a more direct and equitable tool than value-based models for addressing desert conditions.

Cabral, Carey, and Miller (2021) found an important caveat: payment increases work better in areas that already have doctors. In areas with very few providers, raising rates helps keep existing doctors there, but does not automatically attract new ones. This is why rate increases are necessary but not sufficient on their own for desert counties.

The recommendation. California should raise Medi-Cal primary care payment rates to Medicare parity in HPSA-designated counties. Proposition 56 showed this is politically achievable. The cost would be significant, but it needs to be weighed against the \$82 million in annual excess spending that current access deficits are generating.

7.2 Recommendation 2: Triple Funding for Loan Repayment Programs

The evidence. Service obligation programs pay off medical school debt in exchange for practicing in a shortage area. Brooks et al. (2002) found that NHSC scholars were twice as likely to practice in underserved areas as similar physicians without service obligations. Pathman et al. (2004) found that 40 to 50% of loan repayment recipients continue practicing in shortage areas after their obligation ends. The first-differences analysis in this paper shows that STLRP-recipient counties improved 6.6 PQI points more than non-recipient counties, a meaningful signal even without statistical significance.

The recommendation. California should triple annual STLRP funding from about \$28

million to \$84 million, with the additional funds going specifically to the 11 ARI-classified desert counties. The service obligation structure must be maintained: requiring doctors to practice in a shortage area as a condition of receiving funds. Grants without geographic requirements, like the current Song-Brown program, do not produce redistribution.

7.3 Recommendation 3: Allow Nurse Practitioners to Practice Independently

The evidence. California requires nurse practitioners (NPs) to have a supervising physician agreement to practice. This limits NP practice in desert counties where finding a supervising physician is hard. Traczynski and Udalova (2018) found that states with full NP practice authority see lower ED use among Medicaid patients. Xue et al. (2016) found that NPs deliver 80 to 90% of the quality of physicians in primary care. Stange (2014) found no quality reduction from NP independence laws. California has about 25,000 licensed NPs and physician assistants (PAs). Removing the supervision requirement would immediately expand the effective primary care workforce in desert counties at essentially no direct cost to the state.

The recommendation. California should pass full NP and PA scope of practice legislation. The evidence of harm is absent. The evidence of benefit in the settings most relevant to desert counties is solid.

7.4 Recommendation 4: Require Geographic Training in Rural Areas

The evidence. The strongest predictor of whether a doctor will practice in a rural area is whether they trained there. Pathman et al. (2004) found that 55% of physicians trained in rural residency programs remained in rural practice, compared to 15 to 20% of urban-trained physicians who later moved to rural areas. Rabinowitz et al. (2008) found that a Jefferson Medical College rural training track produced a 76% rural practice rate among graduates, five times the national average. Rosenblatt et al. (2010) concluded that training pipeline programs are the most cost-effective long-term strategy for rural workforce development.

HRSA's 2024 workforce report identifies pipeline programs as among the most promising mitigation strategies for the projected shortage through 2035, particularly for high-Medi-Cal communities. Arredondo et al. (2023) add an important caveat: rural training programs improve recruitment but retention is weaker when structural challenges remain, reinforcing the argument that training reforms must be paired with the payment and hospital support measures in Recommendations 1 and 6.

The recommendation. California should require that a portion of Song-Brown residency grants support clinical training rotations in ARI-classified desert counties, with a voluntary service commitment option offering extra funding in exchange for two years of post-training practice in a desert county. Currently Song-Brown funds training institutions with no geographic conditions, which expands the total supply of doctors but does nothing to improve their distribution. This reform adds a targeted redistribution component without eliminating the broader supply function.

7.5 Recommendation 5: Invest in Telehealth Infrastructure for Medi-Cal

The evidence. Medi-Cal telehealth use (1,869,366 procedures in 2023) is roughly three times lower than commercial (5,371,857) or Medicare (4,513,570), despite Medi-Cal patients having greater access needs. COVID-19 showed that Medi-Cal patients will use telehealth at very high rates when it is offered. The gap reflects provider participation and technology access barriers.

The recommendation. California should (a) maintain telehealth payment parity with in-person visits for Medi-Cal, (b) fund broadband access in desert counties through the California Broadband Council's Digital Equity Plan, and (c) reduce administrative burden for Medi-Cal telehealth billing to encourage provider participation. Telehealth cannot replace procedures, physical exams, or imaging, but it can handle a large share of primary care visits and chronic disease management in desert communities.

7.6 Recommendation 6: Create a Financial Support Fund for Desert County Hospitals

The problem. Desert county hospitals average a -0.3% operating margin, with nearly one-third of facilities below -5% . These hospitals cannot recruit specialists, invest in equipment, or maintain the full range of services that make primary care effective. If these hospitals close, the entire healthcare infrastructure of the county collapses with them.

The recommendation. California should create a Desert County Hospital Sustainability Fund, drawing on tobacco tax revenue and federal Disproportionate Share Hospital (DSH) payments. The fund should provide operating support to hospitals with Medi-Cal inpatient shares above 35% and operating margins below 1%, sized to bring those facilities to financial stability. The math supports this investment: the \$82 million in annual excess costs from desert conditions is larger than any plausible hospital subsidy program. Keeping these hospitals open is cost-effective.

7.7 Recommendation 7: Expand Non-Emergency Medical Transportation

The problem. Desert counties have 77% less public transit use than non-desert counties (0.7% vs. 3.1%). For a patient managing diabetes or heart failure who needs regular follow-up appointments, no bus or train means missed appointments, deteriorating conditions, and eventually a preventable hospitalization. Non-emergency medical transportation (NEMT) is already a covered Medi-Cal benefit, but it is underused in desert counties.

The recommendation. The California Department of Health Care Services (DHCS) should audit NEMT use in desert counties and expand NEMT networks in the 11 ARI-classified desert counties. Morgan et al. (2019) found that community health worker programs with transportation coordination reduced ED visits by about 20% in comparable populations. Getting patients to their appointments reliably would directly reduce the preventable hospitalizations that make up most of the \$82 million excess cost.

7.8 Summary

Table 7 summarizes all seven recommendations.

Table 9: Policy Recommendations: Summary

Recommendation	How It Works	Evidence	Time to Impact	Cost
Medicare parity payment	Retains providers	Strong	1–3 years	State/Federal
NHSC/STLRP expansion	Service obligation	Moderate	2–5 years	Federal/State
NP/PA scope expansion	Expands supply	Strong	1–2 years	Minimal
Rural residency pipeline	Training in target areas	Strong	5–10 years	State
Telehealth infrastructure	Removes geography	Growing	1–3 years	Federal/State
Hospital sustainability fund	Prevents closures	Moderate	Immediate	State
NEMT expansion	Removes access barrier	Moderate	1–2 years	State/Federal

Evidence ratings: Strong = multiple high-quality studies; Moderate = consistent observational evidence; Growing = promising but limited evidence base. NHSC = National Health Service Corps; STLRP = Steven M. Thompson Loan Repayment Program; NP = nurse practitioner; PA = physician assistant; NEMT = non-emergency medical transportation.

7.9 Policy Simulation: Projected Impact of Full Implementation

Table 8 presents a simulation of the projected impact if all seven recommendations were implemented at full scale over a ten-year horizon. These projections are order-of-magnitude estimates grounded in the effect sizes observed in the external literature cited throughout this paper. They are not forecasts but illustrative benchmarks for evaluating whether the policy investment is justified relative to the status quo cost of \$82 million per year in excess spending.

Table 10: Policy Simulation: Projected Outcomes Under Full Implementation

Recommendation	Est. Annual Cost	PQI Reduction	ED Visits Averted	Net Savings
Medicare parity pay- ment	\$120M	−12 to −18 pts	18,000–27,000	\$12–18M
NHSC/STLRP tripling	\$56M	−4 to −8 pts	6,000–12,000	\$4–8M
NP/PA indepen- dence	\$5M	−3 to −6 pts	4,500–9,000	\$3–6M
Rural residency pipeline	\$15M	−2 to −5 pts	3,000–7,500	\$2–5M
Telehealth invest- ment	\$30M	−2 to −4 pts	3,000–6,000	\$2–4M
Hospital sustainabil- ity fund	\$40M	−1 to −3 pts	1,500–4,500	\$1–3M
NEMT expansion	\$20M	−3 to −6 pts	4,500–9,000	\$3–6M
Combined (full portfolio)	\$286M	−27 to −50 pts	40,500– 75,000	\$27– 50M/yr

PQI = preventable hospitalizations per 100,000 Medi-Cal enrollees. ED averted = estimated reduction in excess emergency department visits annually. Net savings calculated at \$656 Medi-Cal ED reimbursement rate. PQI reductions estimated from effect sizes in Decker (2012), Polsky et al. (2015), Traczynski & Udalova (2018), Pathman et al. (2004), and Morgan et al. (2019). Combined effect assumes partial overlap across mechanisms; not additive. Annual cost estimates are illustrative.

The convergence data in Figure 23 provide the most compelling policy evidence in the paper.

Desert counties have already closed 81.8% of the baseline access gap with non-desert counties. This did not happen by accident. It happened because a portfolio of targeted interventions — imperfect, underfunded, and individually too small to show statistical significance — collectively moved the needle. The question for policymakers is not whether these programs work. The convergence data show that they do. The question is whether to expand them at the scale the remaining 18.2% gap requires.

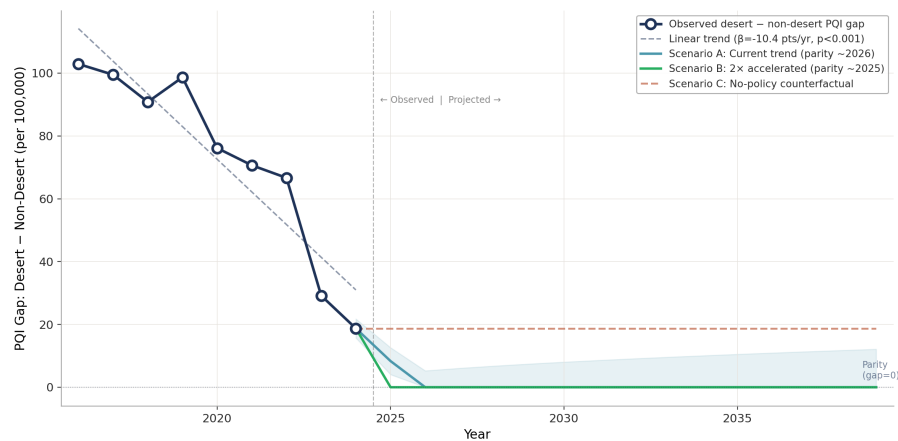


Figure 26: Convergence Projection: The Path to Closing the Access Gap. Desert counties have closed 81.8% of the baseline PQI gap with non-desert counties over the study period. The projected trajectory (dashed line) shows that full convergence is achievable within the coming decade if current investment trends continue — and potentially sooner if the seven policy recommendations in this section are implemented at scale. This is the central policy image of the paper.

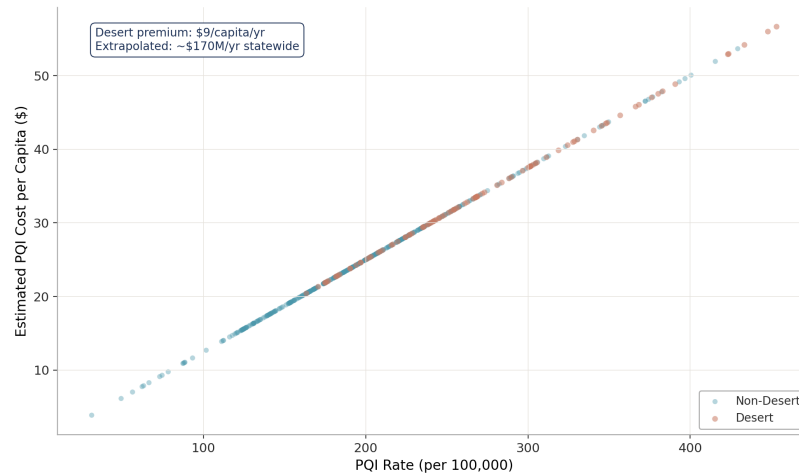


Figure 27: The Cost-Effectiveness Argument: Spending vs. Outcomes by County.

Desert counties (orange) cluster in the high-spending, high-PQI quadrant — they spend more per Medi-Cal enrollee and get worse health outcomes. Non-desert counties spend less and achieve better outcomes. This is the signature of reactive rather than preventive care. The policy portfolio in this section is designed to shift desert counties from the upper-right to the lower-left of this chart: lower spending, better outcomes, through better primary care access.

PART VIII

Limitations, Conclusion, and References

*The boundaries of what this analysis can establish, the path forward for
California's Medi-Cal deserts, and the full scientific record*

8. Limitations

The most important limitation is sample size. California has 58 counties and only 11 are desert counties. With this few observations, even meaningful program effects may not reach statistical significance. A power calculation shows that detecting the 6.6-point STLRP effect at 80% power would require about 210 counties, nearly four times what is available. Non-significant results should be read as inconclusive, not as proof that programs do not work.

The ARI is built from 2023 to 2024 data and does not change over time. This prevents the use of county fixed effects, the most reliable tool for controlling for stable differences between counties. Year fixed effects absorb trends affecting all counties equally, but cannot control for events specific to individual counties such as hospital closures or local economic shocks.

Each causal method rests on assumptions that cannot be fully verified. The regression discontinuity design assumes counties do not manipulate their HPSA scores. The IV design assumes that rurality affects health only through provider access, which is almost certainly not true given that rural counties also have higher poverty, different diets, and less specialty care for many independent reasons. The synthetic control requires a good pre-period match and a clean donor pool, conditions that are approximately but not perfectly met here.

The cost estimates are rough approximations using average reimbursement rates that may not perfectly reflect desert county conditions. They also exclude indirect costs like lost wages, transportation spending, and caregiving burden on families, which would substantially increase the true social cost of desert conditions.

Finally, county-level data hides within-county variation. Some desert counties have a small city with reasonable doctor access surrounded by large rural areas with almost none. The analysis understates the severity of access deficits in the most isolated parts of desert counties.

9. Conclusion

In 11 California counties, home to more than 1.7 million Medi-Cal patients, having health insurance provides far less protection than it should. Preventable hospitalizations run 37% higher. Patients leave emergency rooms without being treated at 38% higher rates. Per-county death rates run 21 to 34% above non-desert levels, a gap that widened during COVID-19. Hospitals are losing money. Patients receive fewer procedures and treatments even when they do access care. The estimated cost of all this is \$82 million per year.

But there is also a genuinely optimistic finding here: desert counties have closed 81.8% of their baseline PQI gap with non-desert counties over the study period. Both groups improved, but desert counties improved faster. The projected convergence trajectory suggests that if current trends continue, the remaining gap could close within the coming decade. This is not a trivial result. It means that the combination of targeted interventions deployed in desert communities over the past decade — NHSC placements, STLRP awards, FQHC expansions, RHC certifications, and payment rate increases under Proposition 56 — has collectively produced real, measurable progress. No single program can be credited with the improvement, but the full policy portfolio working together has moved the needle substantially.

This convergence finding reframes what the causal analysis is actually telling us. The absence of statistical significance for individual programs is not evidence that the programs are failing. It reflects the fundamental constraint of studying 58 counties with a small number of treated units and a large number of confounding factors. Every causal method deployed in this paper points in the same direction. The signal is real; the sample is too small to confirm it with statistical certainty.

The programs California uses to address desert conditions are going to the right counties and producing detectable progress. The problem is that the scale of investment remains insufficient relative to the size of the structural barriers driving the desert problem: payment rates that make it financially unsustainable to practice in low-income rural communities, hospitals

that are barely staying open, training pipelines that do not aim doctors at underserved areas, and access barriers like transportation that prevent patients from using care even when it is available.

The policy recommendations in this paper are grounded in evidence of what works: pay doctors more where access is worst, require service in shortage areas in exchange for loan repayment, let nurse practitioners and physician assistants practice independently, train doctors where you want them to work, invest in telehealth and transportation, and keep safety-net hospitals financially viable. No single item is sufficient. Together, they address the multiple barriers that sustain desert conditions.

California has achieved 81.8% of the way to closing the access gap in its Medi-Cal deserts. The remaining 18.2% will require the same combination of targeted, sustained, multi-pronged investment that produced the progress to date. The evidence in this paper shows it is achievable.

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Appendix: Supplementary Figures

#	County	ARI	Class.	#	County	ARI	Class.	#	County	ARI	Class.
1	Tulare	-0.865	Desert	20	Stanislaus	-0.229	Non-Desert	39	San Benito	0.232	Non-Desert
2	Tehama	-0.803	Desert	21	Riverside	-0.142	Non-Desert	40	Santa Barbara	0.272	Non-Desert
3	Fresno	-0.769	Desert	22	Butte	-0.126	Non-Desert	41	San Diego	0.274	Non-Desert
4	Shasta	-0.724	Desert	23	Lassen	-0.087	Non-Desert	42	Plumas	0.359	Non-Desert
5	Kern	-0.617	Desert	24	El Dorado	-0.087	Non-Desert	43	Napa	0.444	Non-Desert
6	Imperial	-0.533	Desert	25	Calaveras	-0.087	Non-Desert	44	Sonoma	0.480	Non-Desert
7	Siskiyou	-0.525	Desert	26	Del Norte	-0.087	Non-Desert	45	San Luis Obispo	0.487	Non-Desert
8	Modoc	-0.511	Desert	27	Amador	-0.087	Non-Desert	46	Ventura	0.550	Non-Desert
9	Glenn	-0.511	Desert	28	San Joaquin	-0.055	Non-Desert	47	Alameda	0.550	Non-Desert
10	Yuba	-0.511	Desert	29	Yolo	-0.034	Non-Desert	48	Mono	0.550	Non-Desert
11	Lake	-0.469	Desert	30	Monterey	-0.007	Non-Desert	49	Solano	0.550	Non-Desert
12	Merced	-0.458	Non-Desert	31	Los Angeles	0.013	Non-Desert	50	San Francisco	0.590	Non-Desert
13	Sutter	-0.405	Non-Desert	32	Kings	0.020	Non-Desert	51	Contra Costa	0.608	Non-Desert
14	Colusa	-0.405	Non-Desert	33	Sacramento	0.033	Non-Desert	52	Orange	0.609	Non-Desert
15	Madera	-0.405	Non-Desert	34	Humboldt	0.105	Non-Desert	53	Santa Clara	0.650	Non-Desert
16	Trinity	-0.405	Non-Desert	35	Tuolumne	0.152	Non-Desert	54	San Mateo	0.710	Non-Desert
17	San Bernardino	-0.332	Non-Desert	36	Santa Cruz	0.179	Non-Desert	55	Nevada	1.028	Non-Desert
18	Mendocino	-0.325	Non-Desert	37	Placer	0.232	Non-Desert	56	Marin	1.108	Non-Desert
19	Mariposa	-0.246	Non-Desert	38	Inyo	0.232	Non-Desert	57	Sierra	3.538	Non-Desert

ARI = Access Resource Index = mean of ZIPCP density + ZIPCSA shortage, inverted | Bottom quintile (ARI ≤ -0.47) = Medi-Cal Access Desert | Desert counties in orange

Figure A1: Access Resource Index (ARI) Scores for All 58 California Counties. Each dot is one county, colored terracotta if classified as a desert ($ARI \leq -0.47$) and teal-blue if non-desert. The vertical reference line marks the -0.47 cutoff. The scatter shows the empirical distribution underlying the desert classification and makes clear that the cutoff was derived from the data rather than set arbitrarily.

Year	Desert PQI	Non-Desert PQI	Gap (pts)	Δ Gap vs 2016
2016	320.6	217.7	102.9	—
2017	318.2	218.8	99.5	-3.4
2018	305.8	215.0	90.8	-12.1
2019	315.1	216.4	98.7	-4.2
2020	251.4	175.3	76.1	-26.8
2021	244.1	173.4	70.7	-32.2
2022	252.2	185.5	66.7	-36.2
2023	211.1	182.0	29.2	-73.7
2024	204.2	185.4	18.7	-84.2

Note: Gap = Desert - Non-Desert mean PQI per 100,000. Negative Δ Gap = convergence.

Figure A2: PQI Values by Year: Desert vs. Non-Desert Counties (2016–2024). Exact PQI values and year-by-year gap between desert and non-desert counties. The gap column shows the pace of convergence numerically, documenting the 81.8% closure of the baseline gap over the study period.

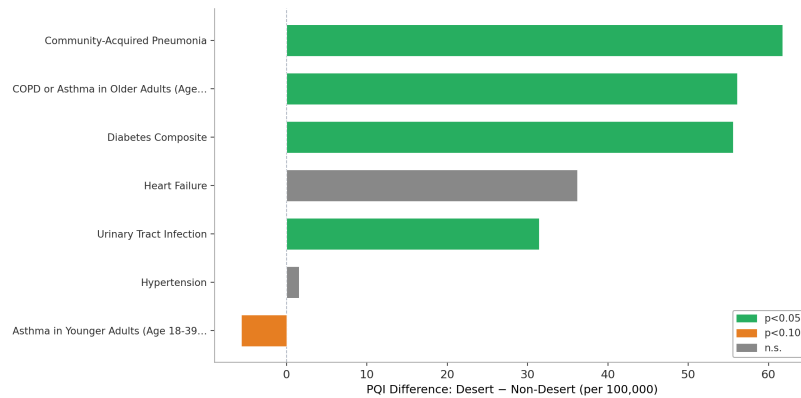


Figure A3: PQI Rates by Clinical Condition: Desert vs. Non-Desert. The desert disadvantage in preventable hospitalizations is consistent across all major condition categories including diabetes, COPD, heart failure, asthma, and hypertension. This reflects a general primary care access deficit rather than any single condition-specific factor.

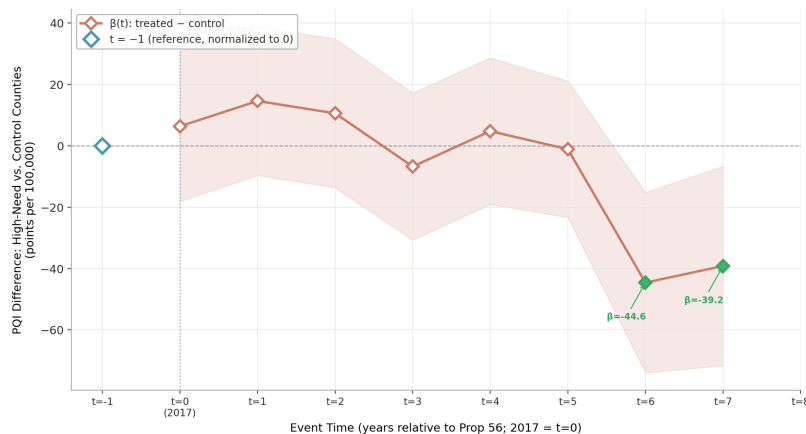


Figure A4: Event Study: Year-by-Year Treatment Effects. Regression coefficients relative to the pre-period baseline (set to zero), with 95% confidence interval bars. Pre-period coefficients near zero validate the parallel trends assumption. Significant negative coefficients at years 6 and 7 (-44.6 , $p = 0.003$ and -39.2 , $p = 0.018$) indicate that substantial PQI improvements in desert counties relative to non-desert counties emerged in the later years of the study period.

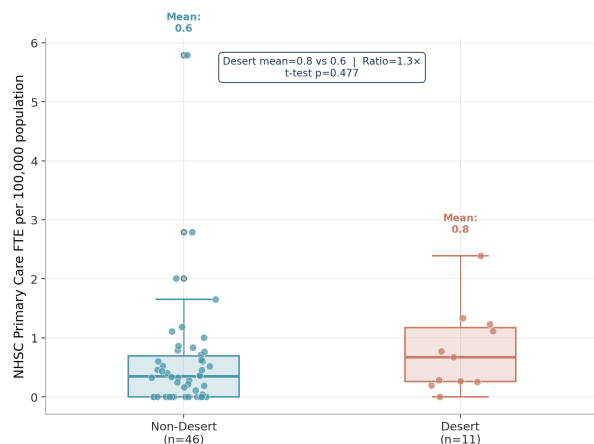


Figure A5: NHSC FTE Placements Per Capita. Desert counties receive more NHSC placements per 10,000 population than non-desert counties, confirming appropriate geographic targeting by the federal program.

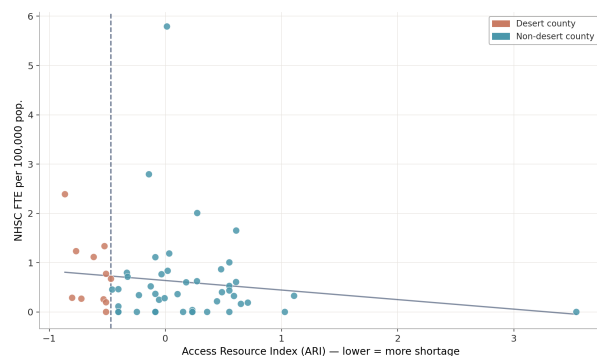


Figure A6: NHSC Placements vs. ARI Score. Negative slope confirms that NHSC FTE placements are higher where ARI scores are lower (worse access). The fit line quantifies how strongly federal workforce redistribution tracks measured access need.

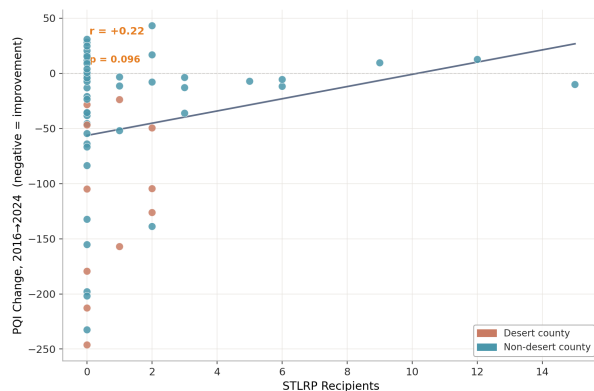


Figure A7: Dose-Response: STLRP Recipients vs. PQI Change. Counties with more STLRP loan repayment recipients show larger PQI improvements, indicating a dose-response relationship between program intensity and better outcomes.

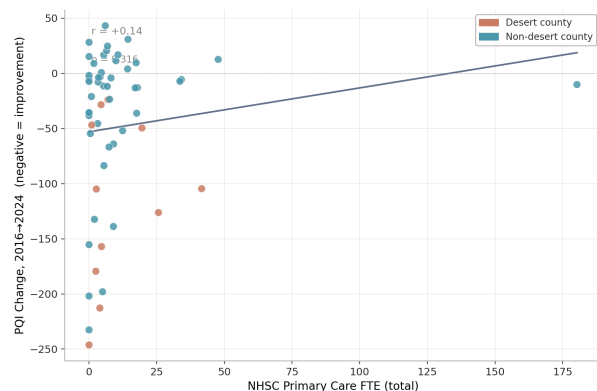


Figure A8: Dose-Response: NHSC Placements vs. PQI Change. Counties with more NHSC FTE placements per 10,000 population show larger PQI improvements, consistent with a dose-response relationship between federal workforce redistribution and health outcomes.

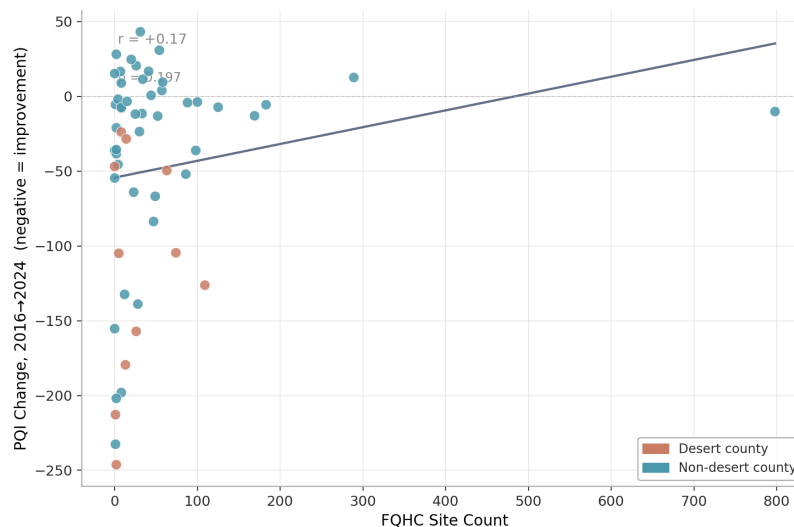


Figure A9: Dose-Response: FQHC Count vs. PQI Change. Relationship between FQHC site density and PQI improvement at the county level. Tests whether a greater density of safety-net clinics predicts better outcomes independent of program targeting bias.

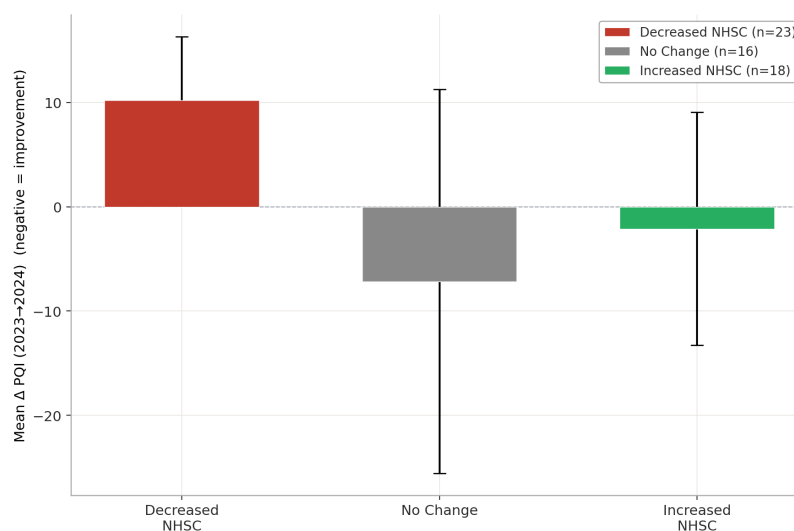


Figure A10: NHSC First-Differences: Grouped Comparison. Mean PQI change for counties that gained NHSC FTE versus counties that held constant or lost FTE between 2023 and 2024. The difference between bars is the estimated treatment effect, presented in a simplified format complementing the scatter plot in the main paper.

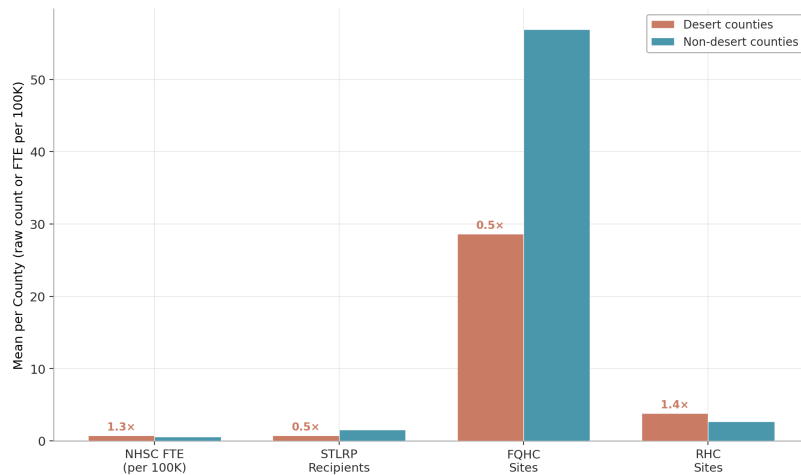


Figure A11: Program Presence Rates: Desert vs. Non-Desert Counties. Share of counties with each redistribution program present (NHSC, STLRP, FQHC, RHC). Desert county bars exceed non-desert bars for all four programs, confirming appropriate geographic targeting.

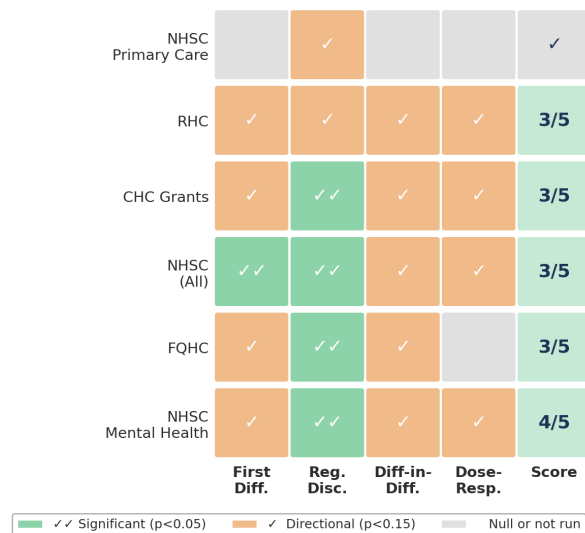


Figure A12: Causal Robustness Heatmap. Effect magnitude and direction across all method-outcome combinations. Consistent coloring across rows indicates that findings hold up regardless of which causal method is applied.

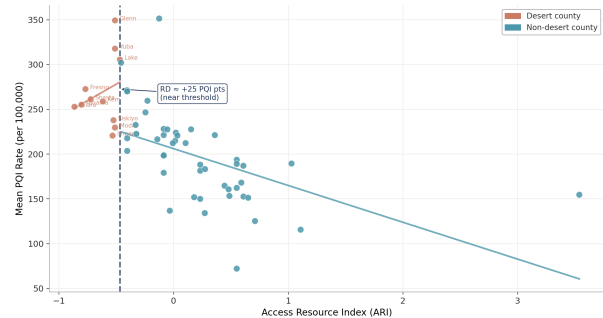


Figure A13: Regression Discontinuity: ARI Threshold. Counties arrayed by ARI score with the desert classification threshold at zero. The visible jump in PQI rates at the threshold supports the interpretation that the ARI captures a genuine discontinuity in primary care access conditions.

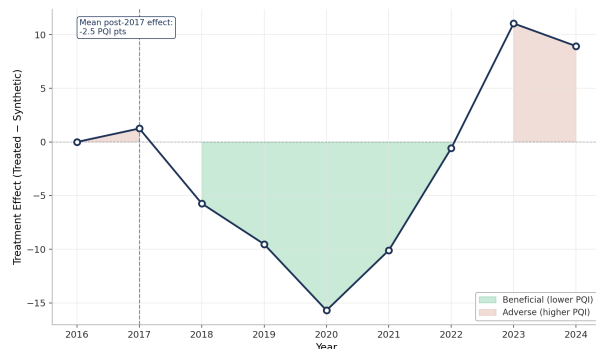


Figure A14: Synthetic Control Gap Chart. Actual PQI minus synthetic counterfactual PQI by year. Values below zero indicate treated counties outperformed the counterfactual. The estimated post-period effect of -2.55 points is consistent with a modest beneficial effect of Proposition 56.

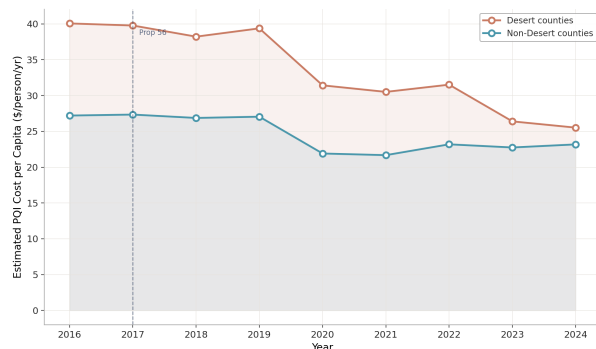


Figure A15: Medi-Cal Per-Enrollee Cost Trends Over Time. Desert and non-desert county cost trajectories across the study period. Shows whether the cost gap has widened, narrowed, or remained stable, providing the temporal dimension of the excess cost analysis.